

# Aircraft Gas Turbine Engine

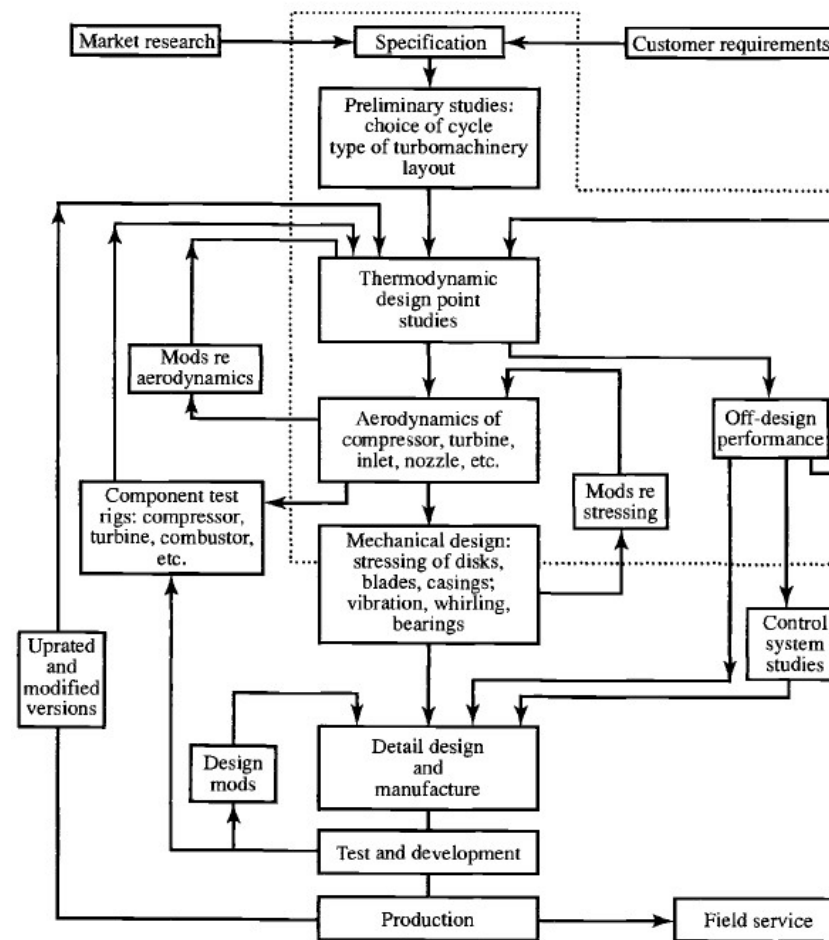
Class 3 – Cycle Analysis

# Introduction

- The following class present the cycle analysis of ideal and real engines performance, we first start our analysis with the basic building block or the common features of any gas turbine engine, namely the gas generator.
- We will then construct a variety of aircraft engines employing the gas generator. These include the turbojet, the afterburning turbojet, the separate and mixed-exhaust turbofan.

# Aircraft Engine Design

- There is no absolute roadmap for the design of a gas turbine engine. The design procedure for a typical gas turbine engine, shown in the picture, requires that thermodynamic design point studies and off design performance be the initial steps of design.



# Brayton Cycle

Aircraft Engine Thermodynamics

# Brayton Cycle

## Ideal Cycle

- Isentropic Compression
- Heat addition @constant pressure.
- Isentropic Expansion
- Heat rejection @constant pressure.

For a calorically perfect gas, thermodynamic analysis of the ideal cycle gives the following equations.

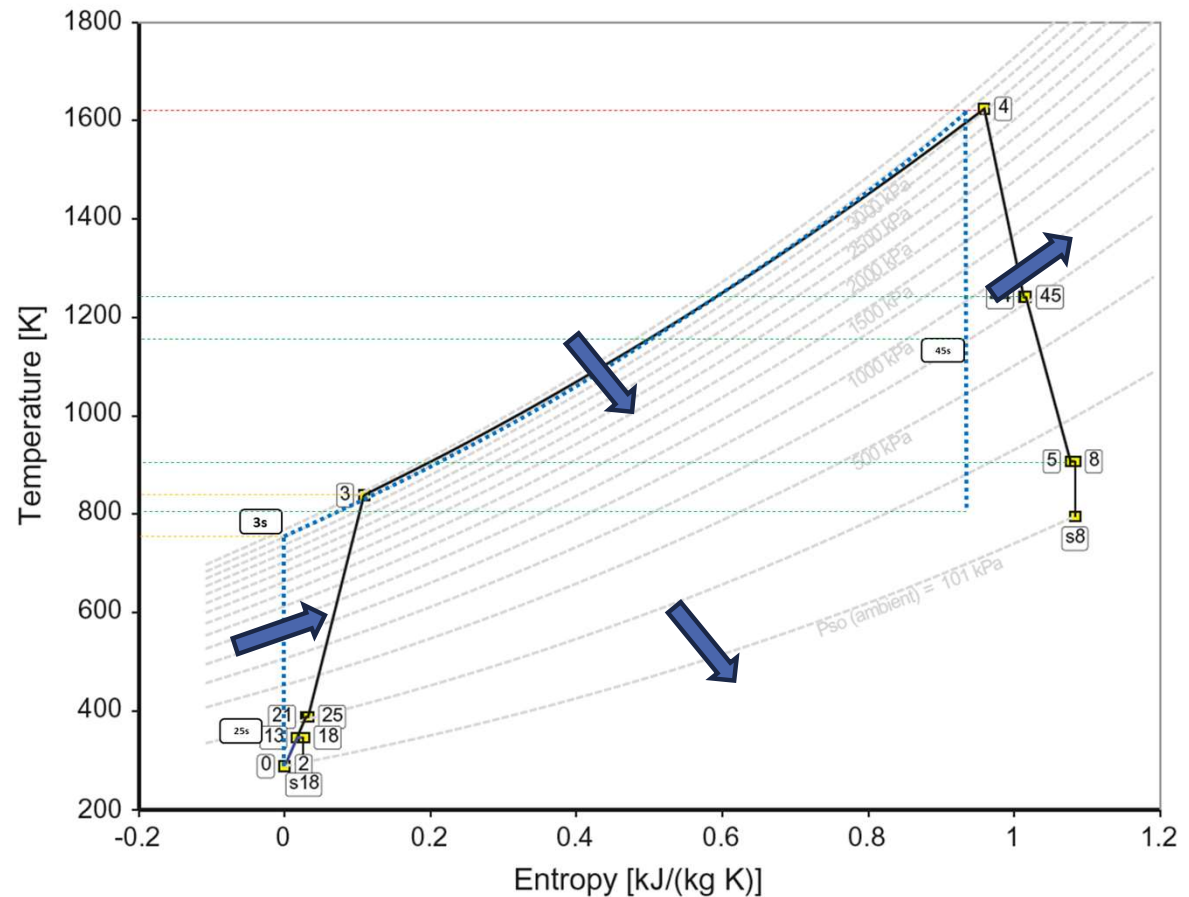
$$\dot{W}_C = \tau_{comp} \omega = \dot{m} C_p (T_2 - T_1)$$

$$\dot{W}_T = \tau_{turb} \omega = \dot{m} C_p (T_3 - T_4)$$

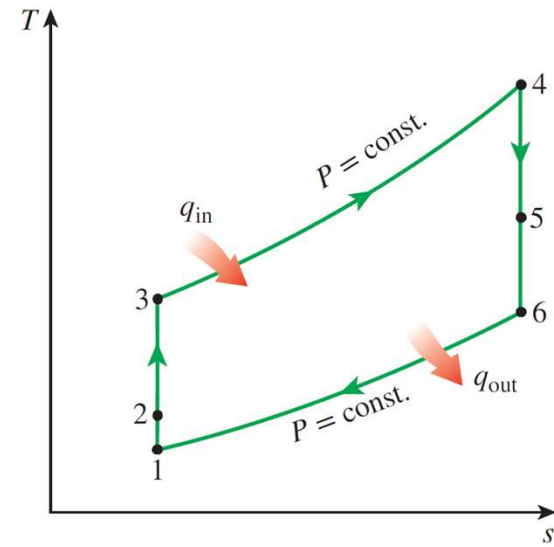
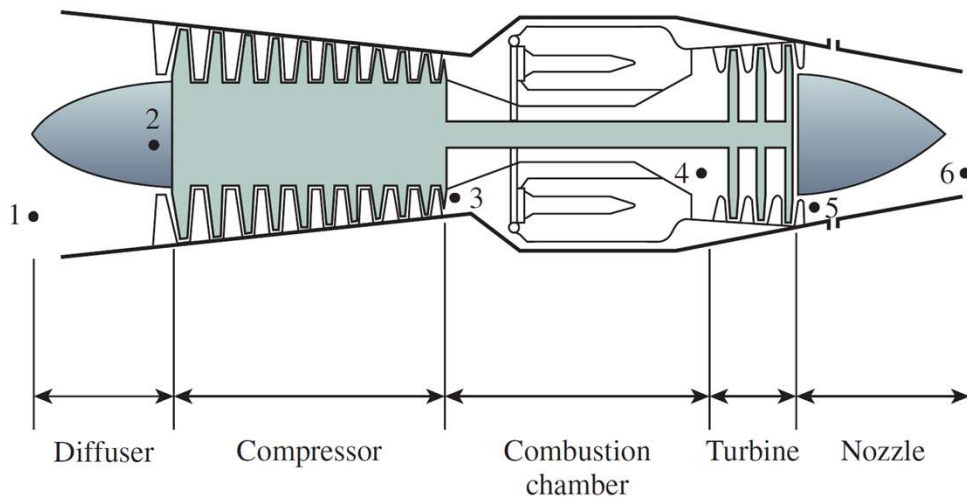
$$Net \dot{W}_{out} = \dot{W}_T - \dot{W}_C$$

Real cycle in a gas turbine differ to the ideal Brayton cycle for different reasons.

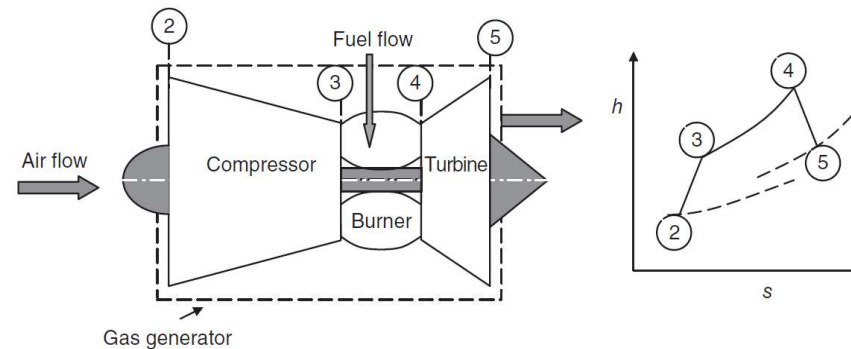
- Pressure losses in the heat addition and rejection are inevitable.
- Work consumption in the cycle will be greater than the work output. This deviation can be calculated by considering the efficiency of both components.



# Brayton Cycle for Aircraft Propulsion



# The Gas Generator

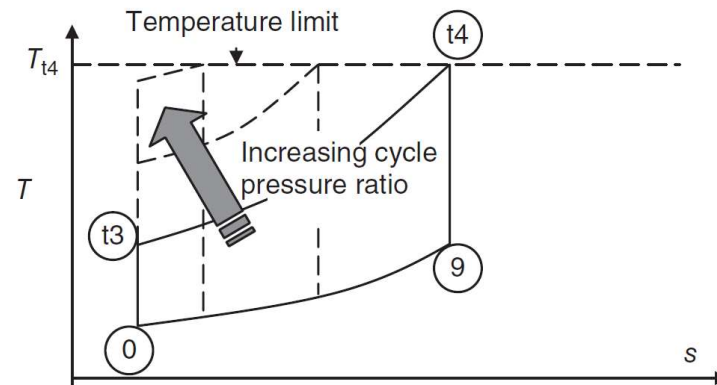


## The Parameters in a Gas Generator

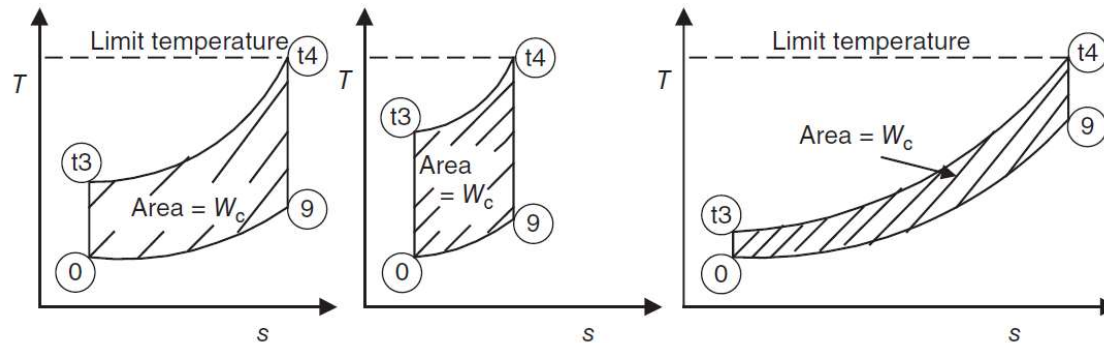
1. Compressor pressure ratio  $\pi_c \equiv \frac{p_{t3}}{p_{t2}}$  ..... (“design” parameter)
2. Compressor air mass flow rate  $\dot{m}_0$  ..... (“size” parameter)
3. Combustor fuel flow rate  $\dot{m}_f$  or turbine entry temperature  $T_{t4}$  ..... (“temperature limit” parameter)
4. Fuel heating value  $Q_R$  ..... (“ideal fuel energy” parameter)
5. Component efficiency ..... (“irreversibility” or loss parameter)

# The Gas Generator

Optimum Pressure Ratio



Optimum Turbine Inlet Temperature





# Engine Main Components

# Inlet/Air Intake

- Its either built into the airframe or it's the forward part of the nacelle installation if the engine is winged or rear fuselage mounted.
- Designed to supply turbulence free air to the first stage of the compressor with the minimum energy loss.

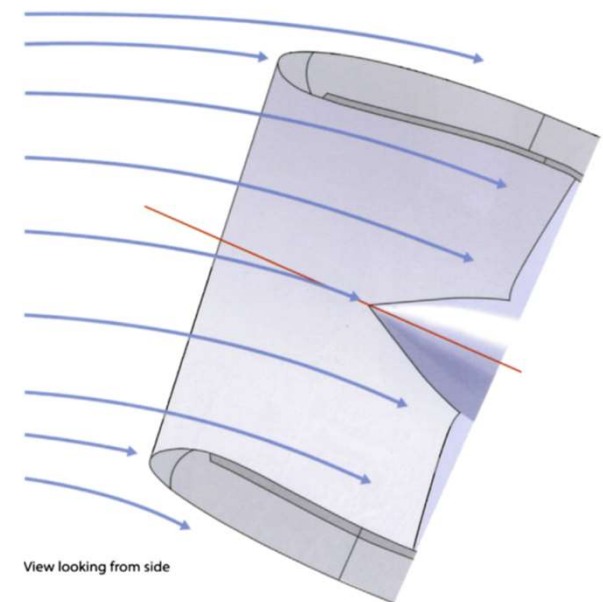
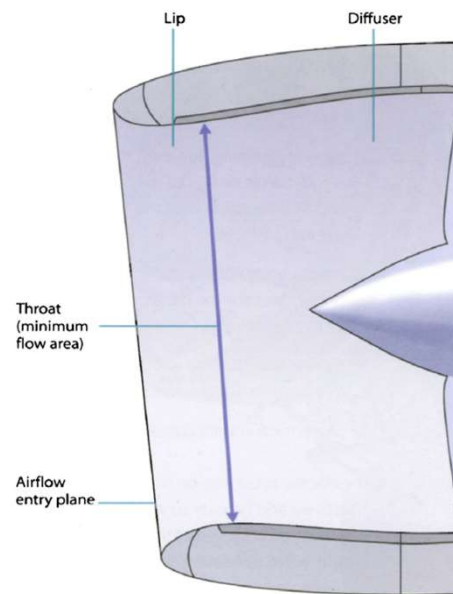


Engine intake



# Inlet/Air Intake

- Main function:
  - Diffuse or decelerate the flow
- Challenge:
  - The adverse pressure gradient c separation



# Inlet thermodynamics

- Inlet adiabatic efficiency

$$\eta_d \equiv \frac{h_{t2s} - h_0}{h_{t2} - h_0} = \frac{(V^2/2)_{ideal}}{V_0^2/2}$$

How else can you represent it?

- Inlet total pressure recovery

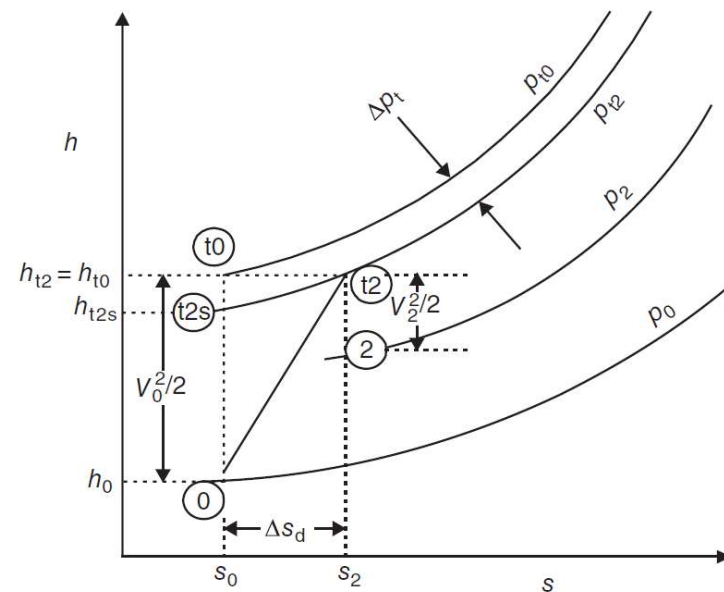
$$\pi_d \equiv \frac{p_{t2}}{p_{t0}}$$

- Entropy rise

$$\frac{p_{t2}}{p_{t0}} = e^{-\frac{s_2 - s_0}{R}}$$

The entropy rise in an adiabatic process leads to a total pressure loss  $\Delta p_t$  following the combined first and second law of thermodynamics

H-S diagram



# Inlet

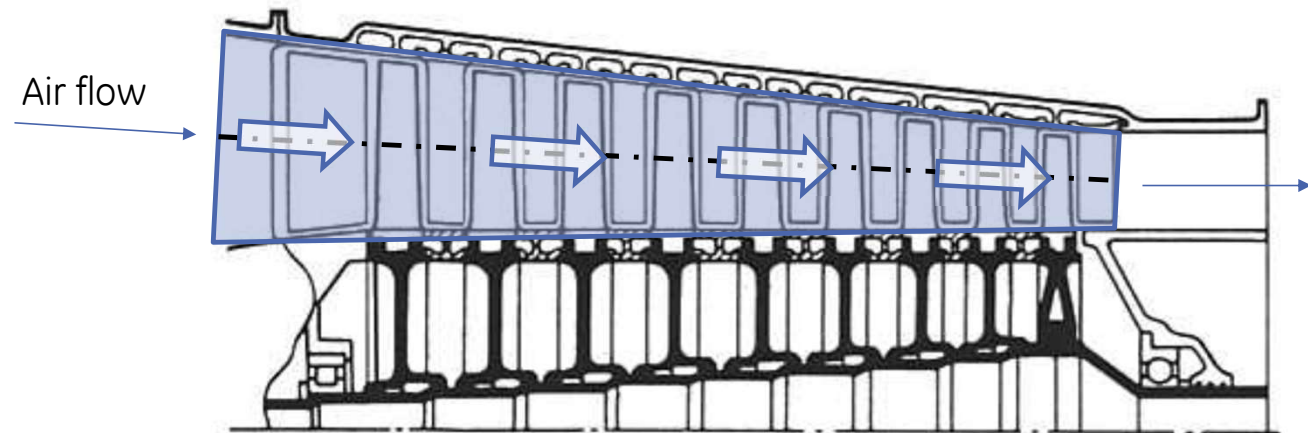
- Important notes:
  - the inlet flow may be considered to be adiabatic, that is,  $h_{t2} = h_{t0}$
  - the inlet flow is always irreversible, that is,  $p_{t2} < p_{t0}$ , with viscous dissipation in the boundary layer and in a shock as the sources of irreversibility ( $s_2 > s_0$ )
  - there are two figures of merit that describe the extent of losses in the inlet and these are  $\eta_d$  and  $\pi_d$
  - the two figures of merit are related via nozzle pressure recovery
  - in cruise,  $A_0 < A_2$  and hence a diffusing passage and at low speed or takeoff,  $A_0 > A_2$ , that is, a nozzle
  - the outer nacelle geometry of an inlet dictates the drag divergence and high angle of attack characteristics of the inlet and is crucial for the installed performance.

## Example 4.1

- An aircraft is flying at an altitude where the ambient static pressure is  $p_0 = 10$  kPa and the flight Mach number is  $M_0 = 0.85$ . The total pressure at the engine face is measured to be  $p_{t2} = 15.88$  kPa. Assuming the inlet is adiabatic and  $\gamma = 1.4$ , calculate
  - a) the inlet total pressure recovery  $\pi$
  - b) the inlet adiabatic efficiency  $\eta$
  - c) the nondimensional entropy rise caused by the inlet  $\Delta s/R$

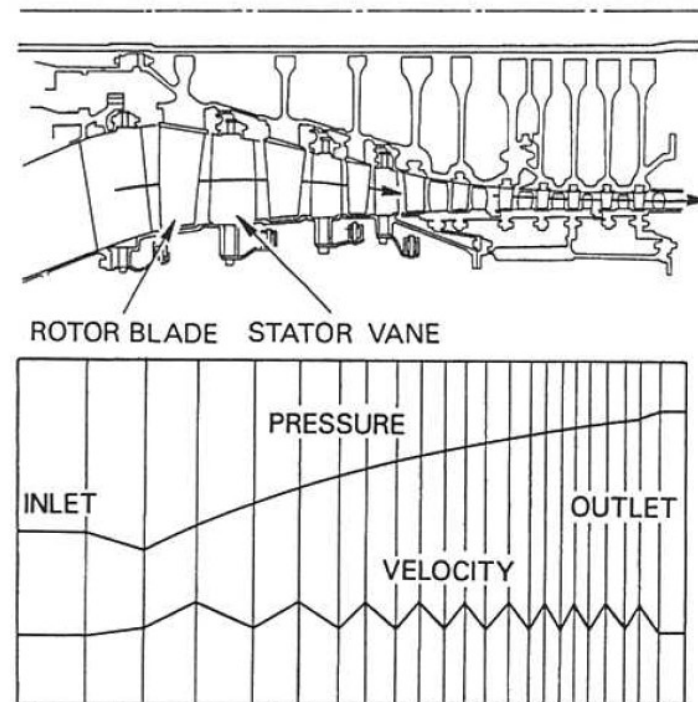
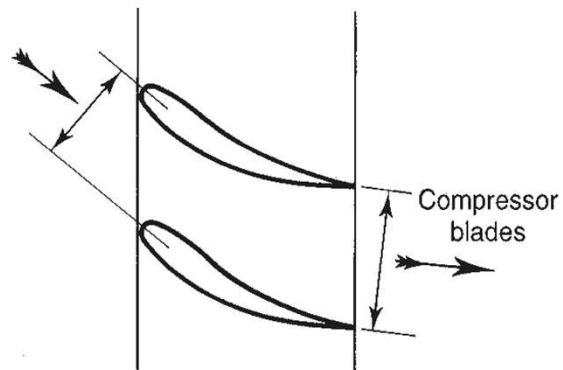
# Compressor

- The function of a compressor is to take a large quantity of atmospheric air and raise its pressure before delivering it to the combustor. The compressor encloses alternating rows of rotating and fixed blades.
- One rotating blade followed by one stationary vane is called a stage.
- The flow path of an axial compressor decreases in volume in the direction of the air flow, thus increasing air pressure, while maintaining the air mass flow rate.



# Compressor

- Challenges?





# Compressor Thermodynamics

## H-S diagram

- Inlet adiabatic efficiency

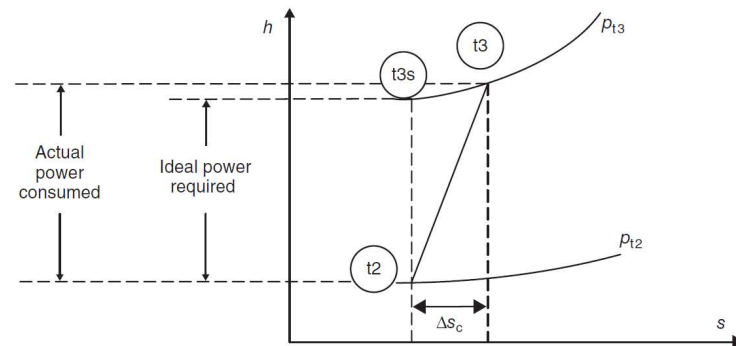
$$\eta_c \equiv \frac{h_{t3s} - h_2}{h_{t3} - h_2} = \frac{\Delta h_{t,isentropic}}{\Delta h_{t,actual}}$$

How else can you represent it?

- Polytropic efficiency

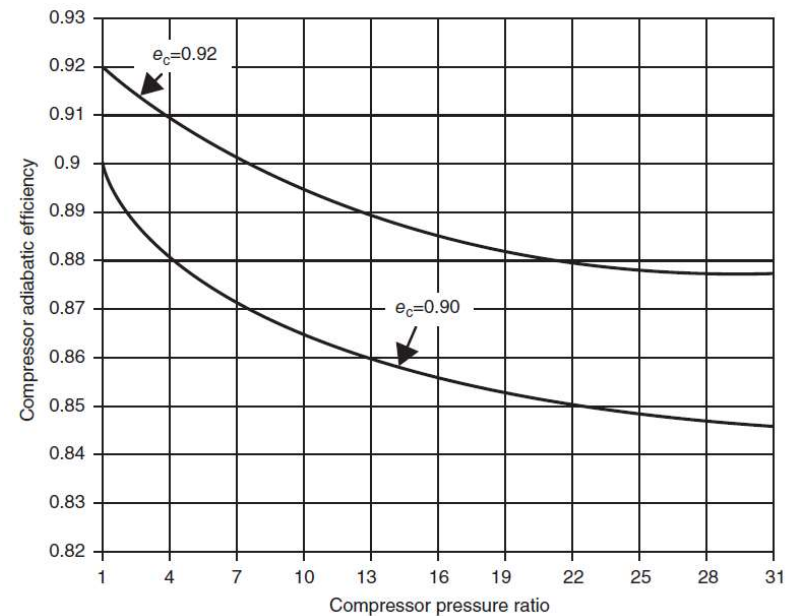
$$e_c = \frac{\frac{dp_t}{p_t}}{\frac{C_p dT_T}{RT_t}}$$

How can we relate both?

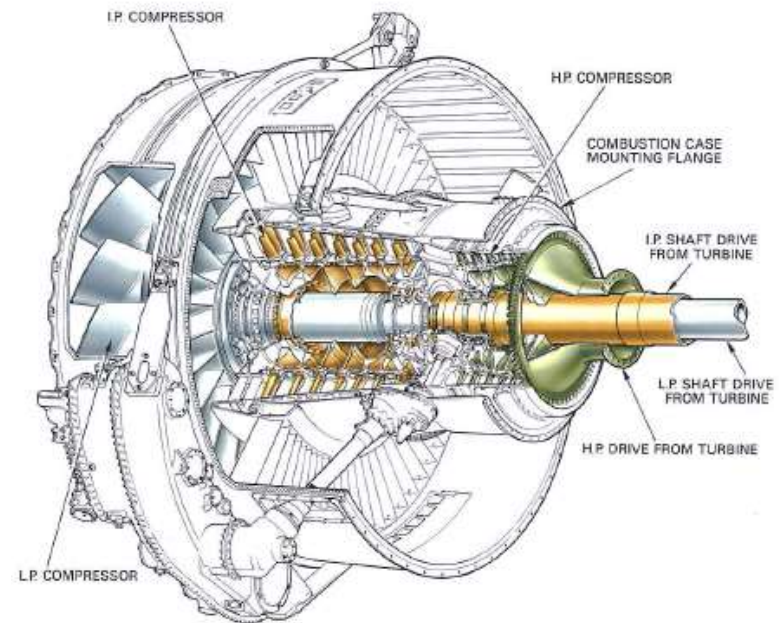
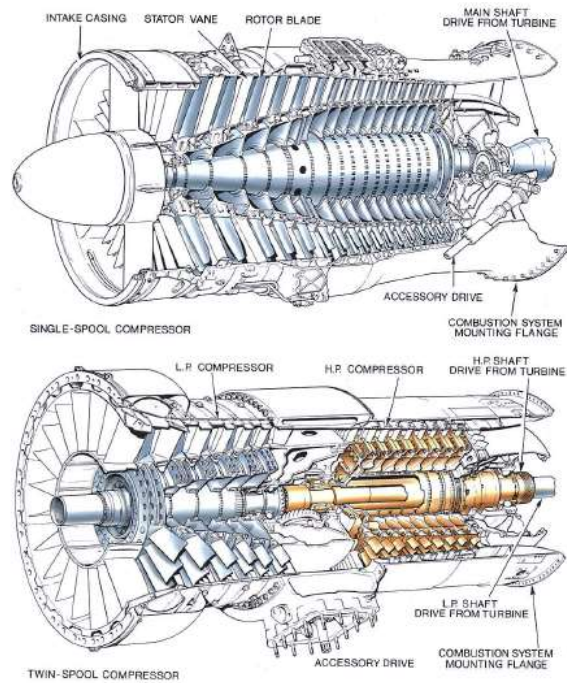


# Compressor

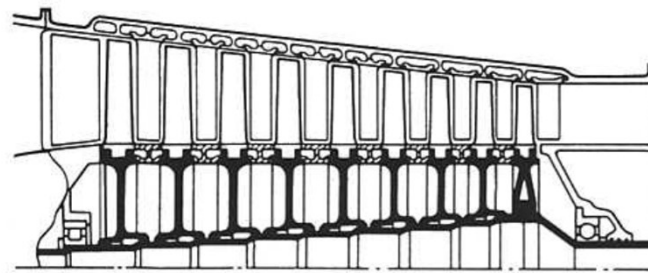
- Important Notes:
  - a real compressor flow may be considered adiabatic, that is,  $\dot{Q}_{\text{compressor}} \approx 0$
  - the energy transfer to the fluid due to the shaft in a compressor is several orders of magnitude higher than any heat transfer that takes place through the casing, thus heat transfer is neglected
  - viscous dissipation in the wall boundary layer and shocks account for the sources
  - of irreversibility in a compressor there are two figures of merit that describe the compressor efficiency, one is
  - the adiabatic compressor efficiency  $\eta_c$  (sometimes referred to as "isentropic" efficiency) and the second is the polytropic or small-stage efficiency  $e_c$
  - the two compressor efficiencies are interrelated, that is,  $\eta_c = \eta_c(\pi_c, e_c)$
  - the compressor polytropic efficiency is independent of compressor pressure ratio  $\pi_c$
  - the compressor adiabatic efficiency is a function of  $\pi_c$  and decreases with increasing pressure ratio
  - to achieve a high-pressure ratio in a compressor, multistage and multispool configurations are needed
  - in a gas turbine engine, the compressor power is derived from a shaft that is connected to a turbine.



# Compressor Arrangements



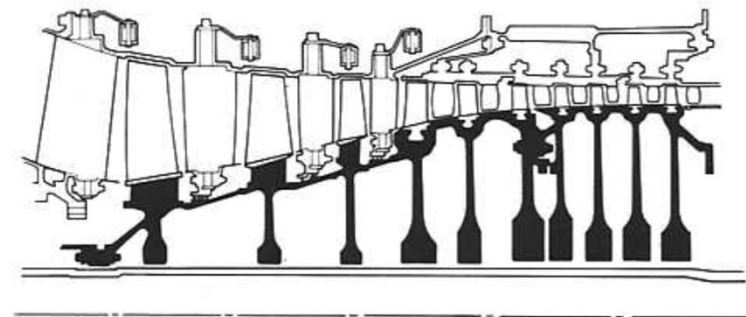
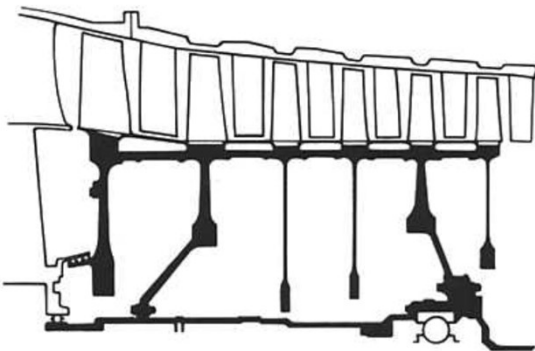
# Compressor Arrangements



Constant Mean Radius

Constant hub radius

Constant Tip Radius

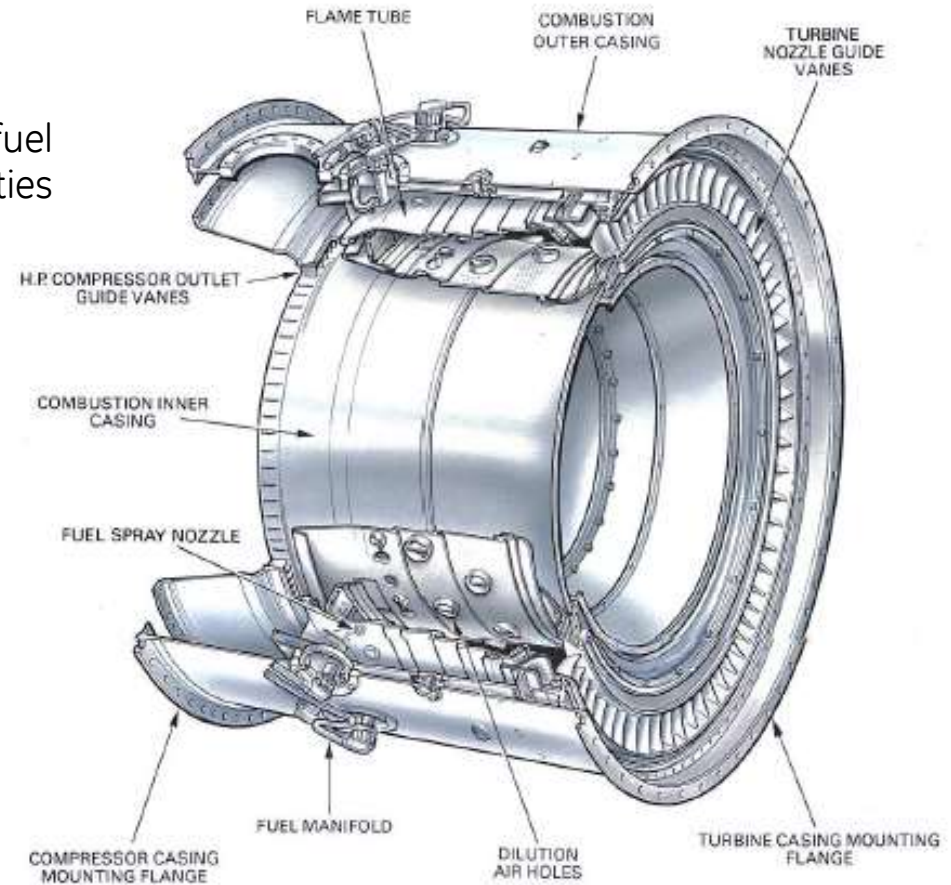


## Example 4.2

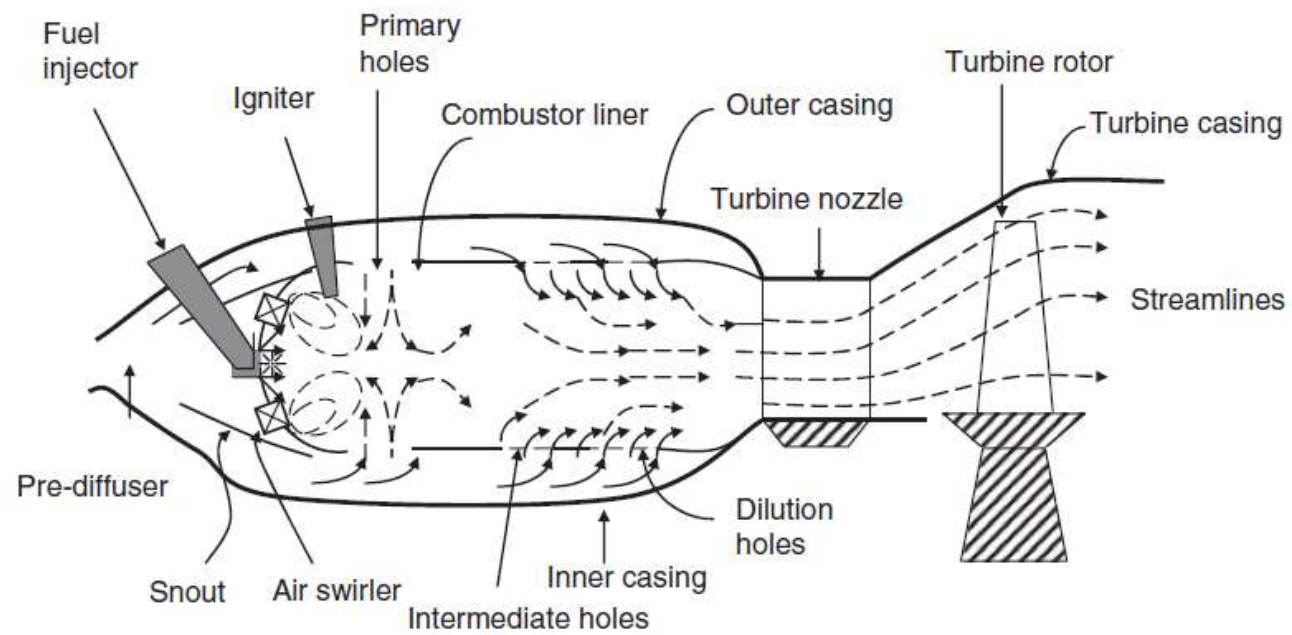
- A multistage axial-flow compressor has a mass flow rate of 50 kg/s and a total pressure ratio of 35. The compressor polytropic efficiency is  $\epsilon_c = 0.90$ . The inlet flow condition to the compressor is described by  $T_2 = 288$  K and  $p_2 = 100$  kPa. Assuming the flow in the compressor is adiabatic, and constant gas properties throughout the compressor are assumed, i.e.,  $\gamma = 1.4$  and  $c_p = 1004$  J/kg  $\cdot$  K, calculate
  - a) compressor exit total temperature  $T_3$  in K
  - b) compressor adiabatic efficiency  $\eta_c$
  - c) compressor shaft power  $W_c$  in MW

# Combustor

The combustor burns large quantities of fuel through small nozzles with large quantities of air supplied by the compressor.



# Combustor



# Combustor thermodynamics

- Burner Efficiency

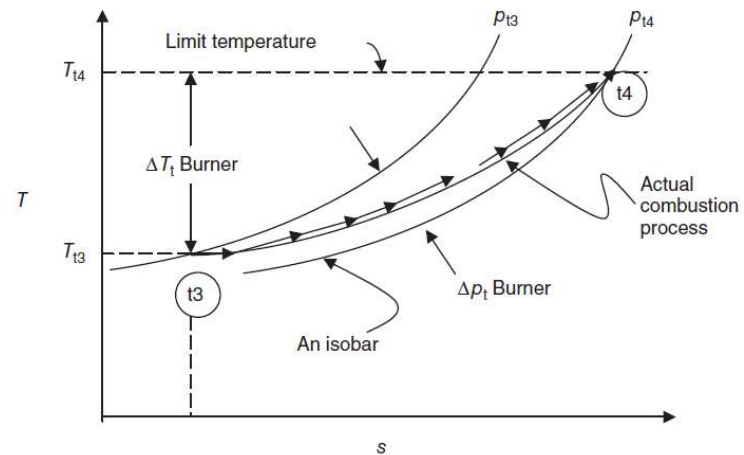
$$\eta_b = \frac{(\dot{m}_3 + \dot{m}_f)h_{t4} - \dot{m}_3h_{t3}}{\dot{m}_f(Q_r)}$$

Challenges or relevance?

- Important notes:

- the fuel is characterized by its heating value  $Q_r$  (maximum releasable thermal energy per unit mass)
- the burner is characterized by its efficiency  $\eta_b$ , and its total pressure ratio  $\pi_b$
- burning at finite Mach number, frictional losses on the walls and turbulent mixing are identified as the sources of irreversibility, that is, losses, in a burner
- the fuel-to-air ratio  $f$  and the burner exit temperature  $T_{t4}$  are the thrust control/engine design parameters
- the application of the energy balance across the burner yields either  $f$  or  $T_{t4}$

H-S Diagram



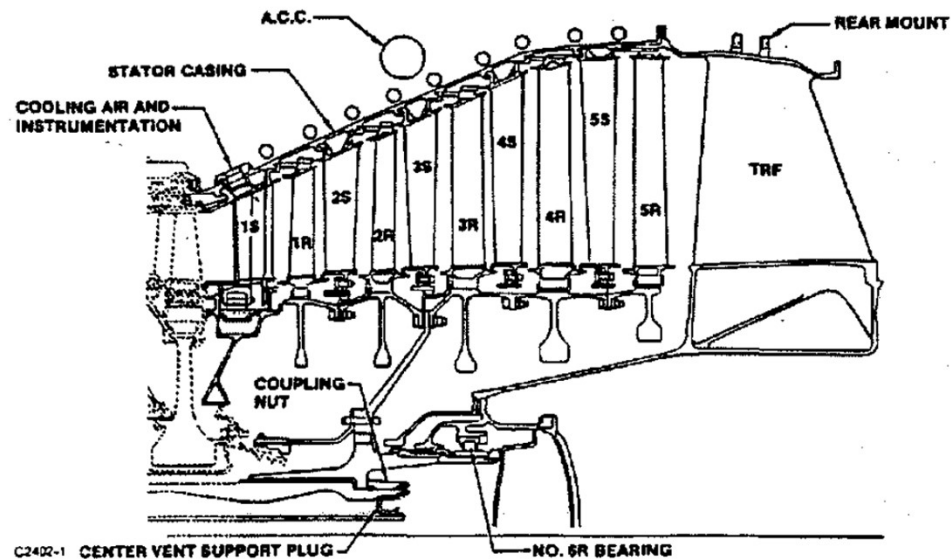


## Example 4.3

- A gas turbine combustor has inlet condition  $T_3 = 800$  K,  $p_3 = 2$  MPa, air mass flow rate of 50 kg/s,  $\gamma_3 = 1.4$ ,  $c_{p3} = 1004$  J/kg · K. A hydrocarbon fuel with ideal heating value  $Q_R = 42,000$  kJ/kg is injected in the combustor at a rate of 1 kg/s. The burner efficiency is  $\eta_b = 0.995$  and the total pressure at the combustor exit is 96% of the inlet total pressure, i.e., combustion causes a 4% loss in total pressure. The gas properties at the combustor exit are  $\gamma_4 = 1.33$  and  $c_{p4} = 1156$  J/kg · K. Calculate
  - a) fuel-to-air ratio  $f$
  - b) combustor exit temperature  $T_4$  in K and  $p_4$  in MPa

# Turbine

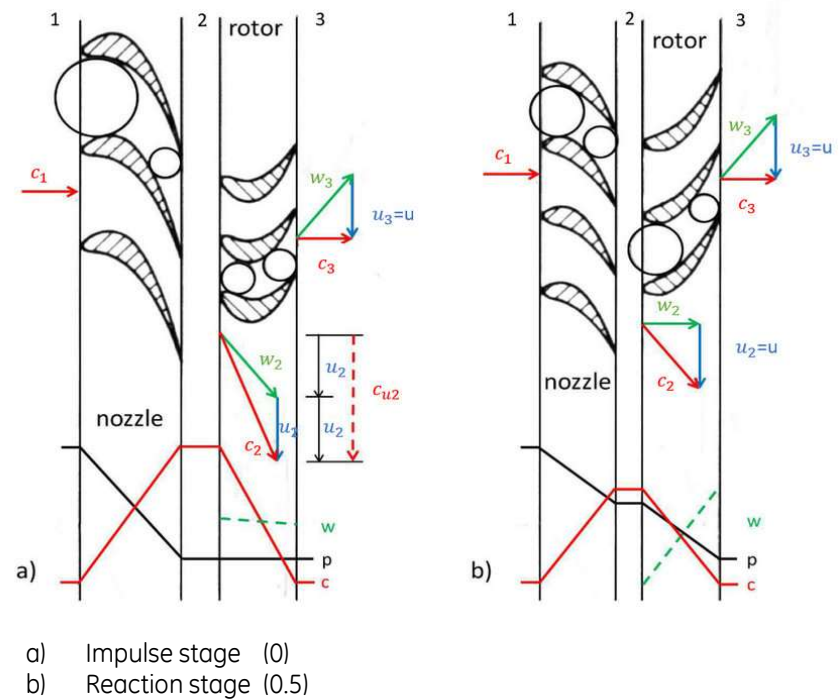
- The function of the turbine cylinder section is to extract useful work from the hot gas generated in the combustor. The gases expand through the sequential turbine stages that convert thermal energy to mechanical energy.



Five - stage LPT Flowpath (CF6-80C2)

# Turbine

- The high pressure and temperature gas that leaves the combustor is directed into a turbine. The turbine may be thought of as a valve because on one side it has a high-pressure gas and on the other side it has a low-pressure gas of the exhaust nozzle or the tailpipe.
- Challenges?

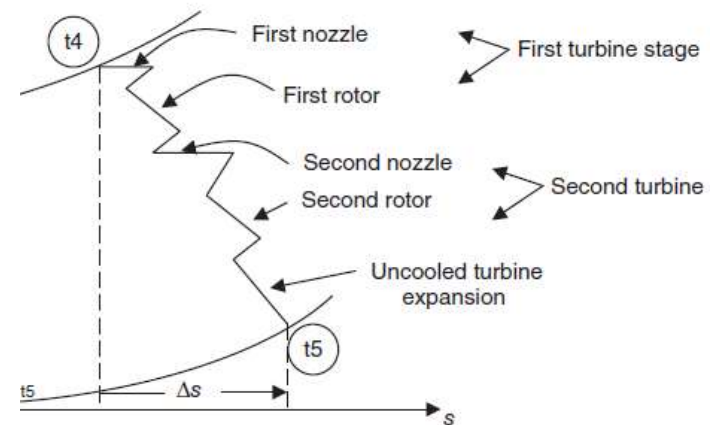
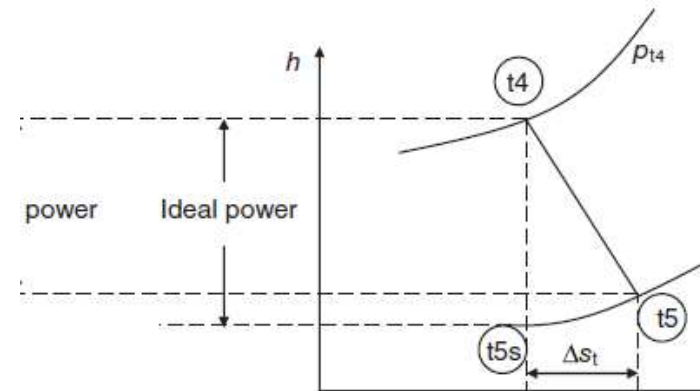
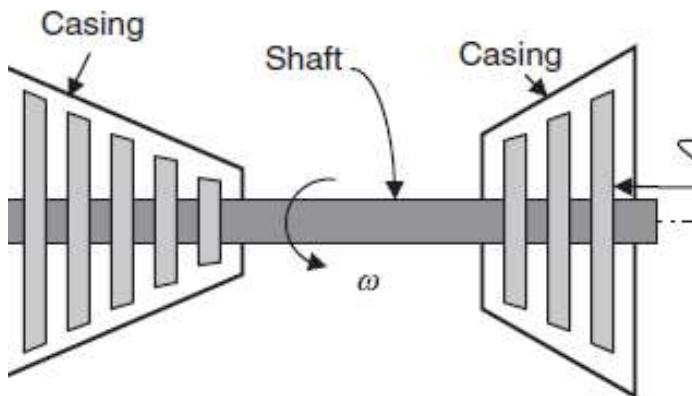


# Turbine Thermodynamics

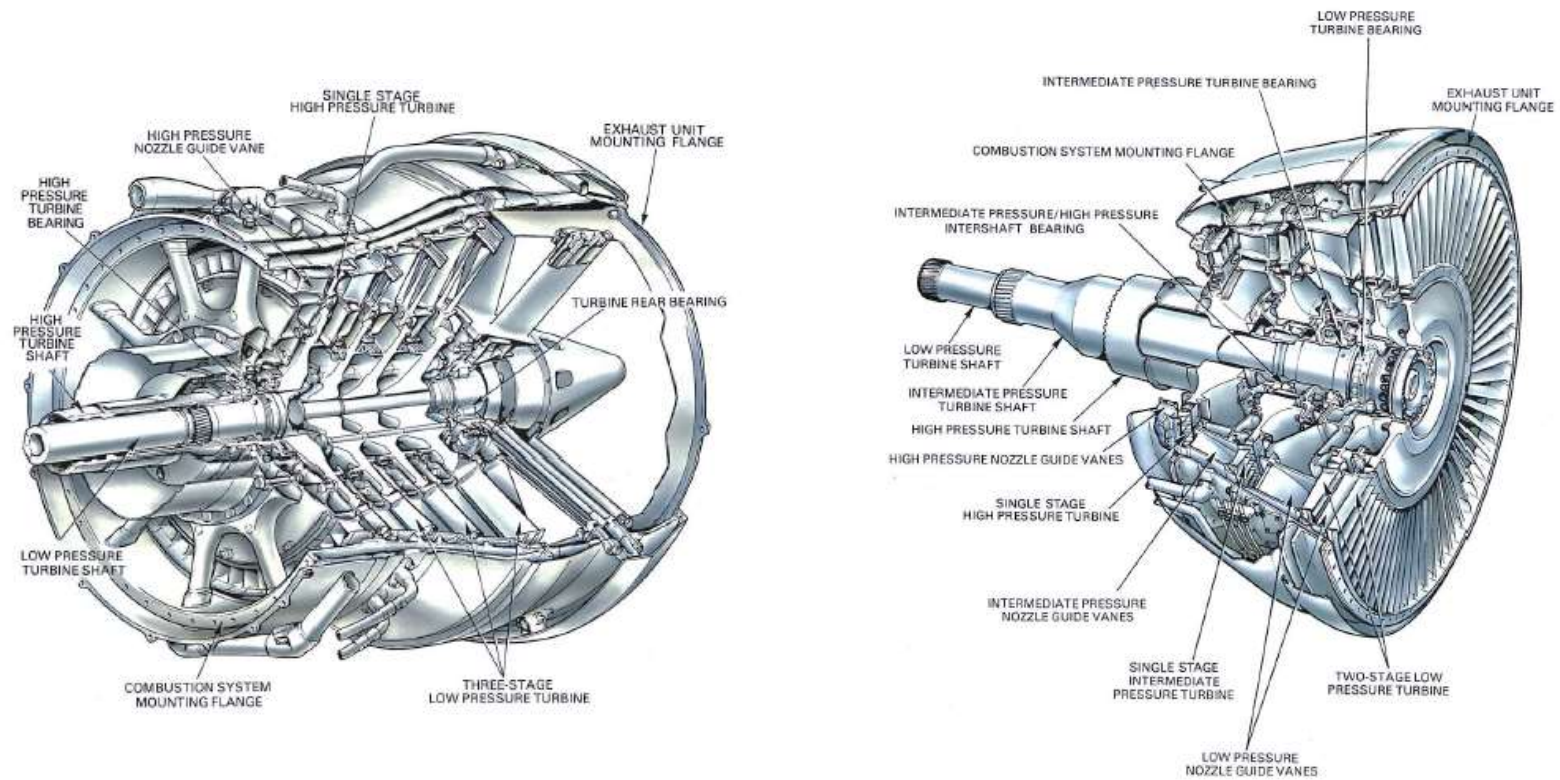
- Turbine adiabatic efficiency

$$\eta_t \equiv \frac{h_{t4} - h_{t5}}{h_{t4} - h_{t5s}} = \frac{\Delta h_{t,actual}}{\Delta h_{t,isentropic}}$$

What about polytropic efficiency?

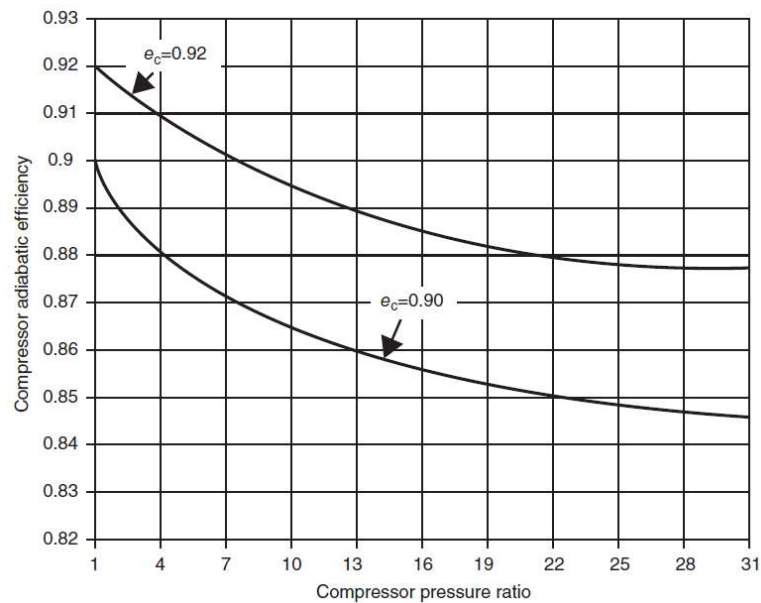


# Turbine Arrangements

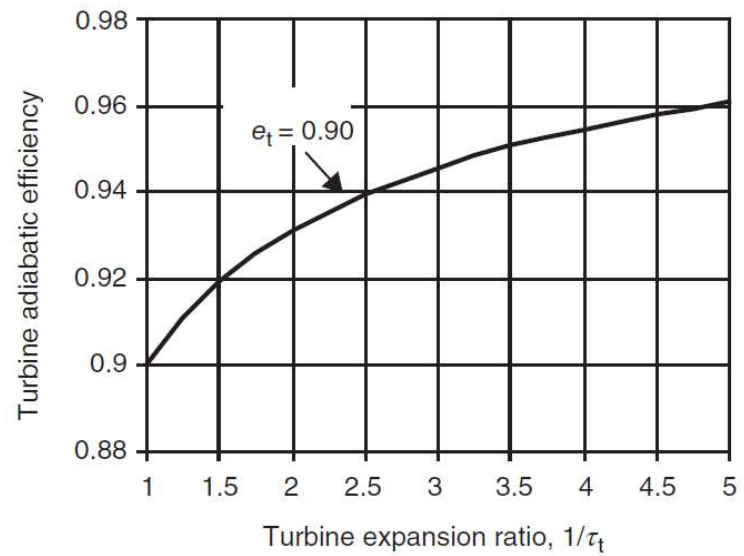


# Polytropic Efficiency

Compressor

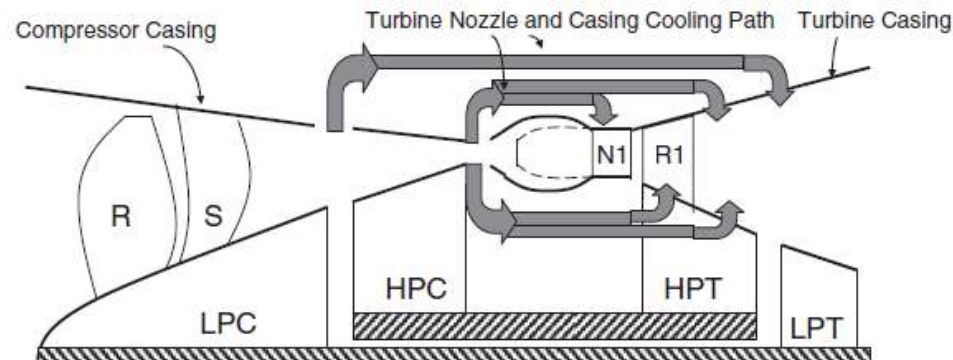


Turbine



# Turbine Blade Cooling

- To cool the high-pressure turbine stages, a small fraction of compressor air can be diverted from various stages of the compression. Engine cooling is essentially a pressure driven process, which calls for a pressure scheduling of the coolant to achieve the highest cooling efficiency.



# Turbine

In summary we learned that

- the flow expansion in the turbine produces the needed shaft power for the compressor and other propulsion system or aircraft needs, for example, an electric generator
- there are two figures of merit in a turbine, which measure the extent of irreversibility in a turbine,  $\eta_t$  and  $\epsilon_t$  and they are related
- the gas expansion in an uncooled turbine is treated as adiabatic
- the frictional losses on the blades and the casing as well as any shock losses, in relative supersonic passages, are the sources of irreversibility in an uncooled turbine
- turbulent mixing losses between the coolant and the hot gas is an added source of irreversibility in a cooled turbine
- the turbine entry temperature  $T_4$  is a design parameter that sets the stage for the turbine material and cooling requirements
- the power balance between the turbine, the compressor, and other known power drainage, establishes the turbine expansion ratio  $\pi_t$
- cooling of the high-pressure turbine is achieved through compressor air bleed that is injected through the blade root in the rotor and the casing for the turbine nozzle
- the turbine nozzle is choked (i.e., the throat Mach number is 1), over a wide operating range of the engine, and as such is the first *throttle station* of the engine.



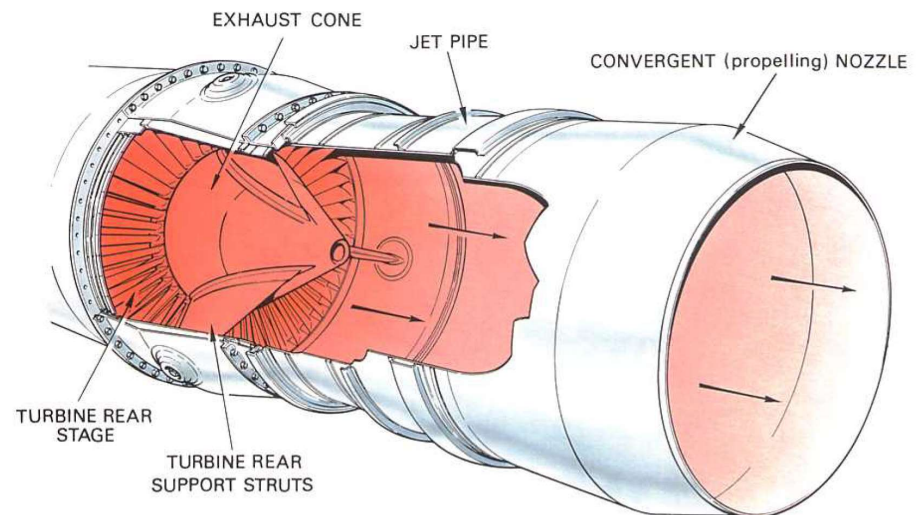
## Example 4.4

- Consider an uncooled gas turbine with its inlet condition the same as the exit condition of the combustor described in Example 4.3. The turbine adiabatic efficiency is 88%. The turbine produces a shaft power to drive the compressor and other accessories at  $\dot{\phi}_t = 45$  MW. Assuming that the gas properties in the turbine are the same as the burner exit in Example 4.3, calculate
  - a) turbine exit total temperature  $T_5$  in K
  - b) turbine polytropic efficiency,  $\eta$
  - c) turbine exit total pressure  $p_5$  in kPa
  - d) turbine shaft power  $\dot{\phi}_t$  based on turbine expansion  $\Delta T$

# Nozzle

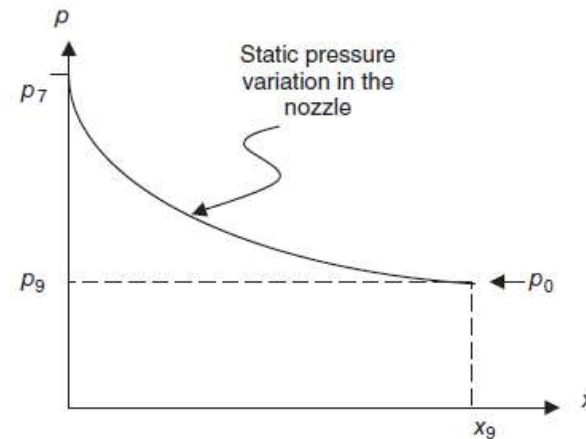
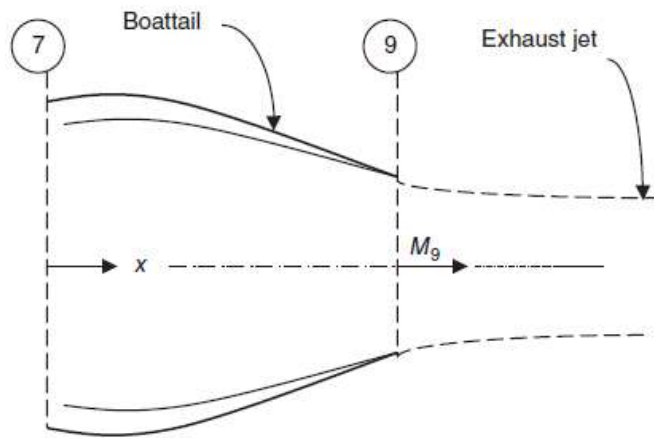
- The primary function of an aircraft engine exhaust system is to accelerate the gas efficiently. The nozzle parameter that is of utmost importance in *propulsion* is the gross thrust,  $F_G$ . The expression derived earlier for the gross thrust was

$$F_G = (\dot{m}_0 + \dot{m}_f)V_9 + (p_9 - p_0)A_9$$



# Nozzle

- As the fluid accelerates in a nozzle, the static pressure drops and hence a *favorable pressure gradient* environment is produced in the nozzle. This is in contrast to diffuser flows where an *adverse pressure gradient* environment prevails.



# Nozzle Thermodynamics

- Rules:

1) If  $M_{jet} < 1$ , then  $p_{jet} = p_{ambient}$ .

A convergent duct causes flow acceleration in a subsonic stream to a maximum Mach number of 1, which can only be reached at the minimum area of the duct, namely at its exit. So we stipulate that for all subsonic jets, that is,  $M_9 < 1$ , there is a static pressure equilibrium in the exhaust stream and the ambient fluid, that is,  $p_9 = p_0$ .

2) If  $p_{jet} = p_{ambient}$  then we have a perfectly expanded nozzle which results in max thrust.

3) If  $NPR \geq (NPR)_{critical}$  then the nozzle throat velocity is Sonic (i.e., choked),  $M_8 = 1.0$

A real nozzle flow experiences total pressure loss due to viscous dissipation in the boundary layer as well as shock waves, and yet it is possible for it to perfectly expand the gas to the ambient condition. The exit pressure  $p_9$  is a direct function of the nozzle area ratio  $A_{exit}/A_{throat}$ , and the nozzle pressure ratio (NPR).

4) If  $p_{t7}/p_0 \geq \sim 2$  then the nozzle throat can be choked, i.e.  $M_8 = 1.0$

$$NPR \equiv \frac{p_{t7}}{p_0}$$

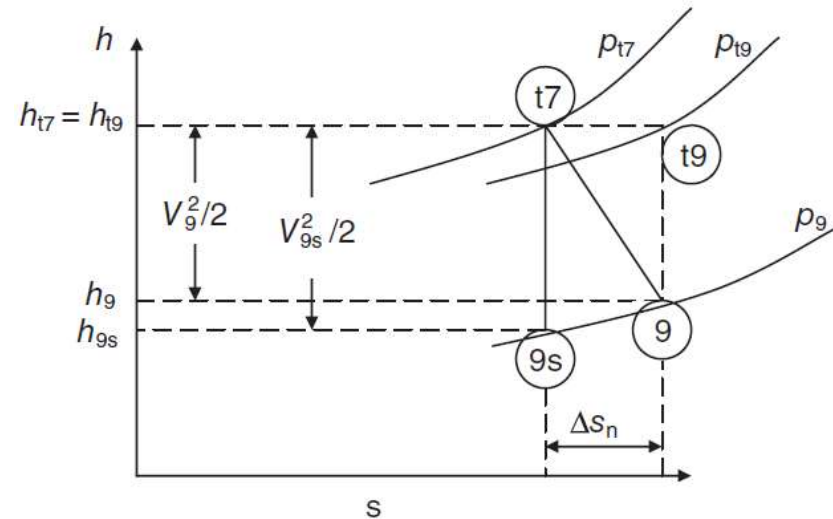
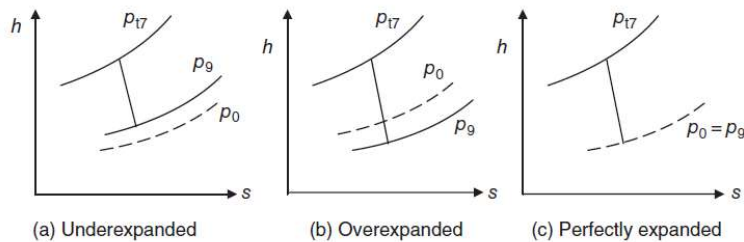
5) Static pressure must be continuous across the slipstream of a jet exhaust plume

$$h_{t7} = h_{t9}$$

# Nozzle Thermodynamics

- Nozzle Isentropic Efficiency

$$\eta_n \equiv \frac{h_{t7} - h_9}{h_{t7} - h_{9s}}$$

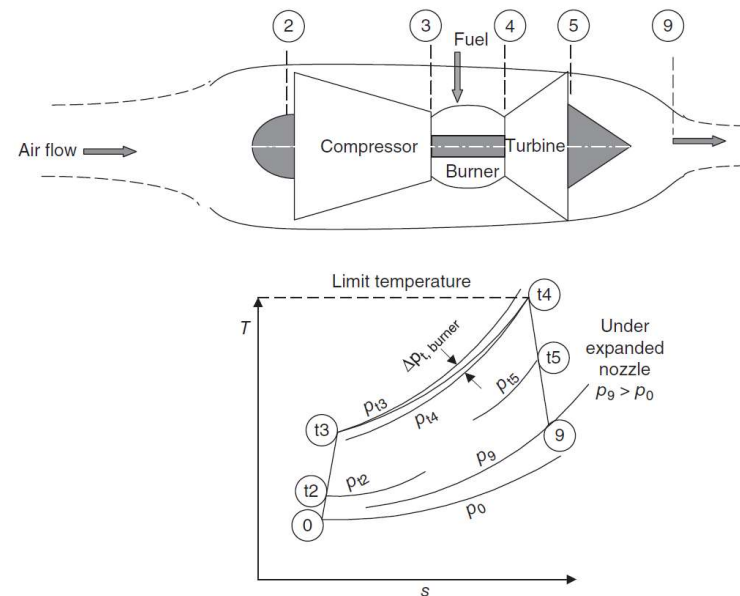


## Example 4.7

- Consider a convergent-divergent nozzle with a pressure ratio  $NPR = 10$ . The gas properties are  $\gamma = 1.33$  and  $c_p = 1,156 \text{ J/kg} \cdot \text{K}$  and remain constant in the nozzle. The nozzle adiabatic efficiency is  $\eta_n = 0.94$ . Calculate
  - a) nozzle total pressure ratio  $\pi_n$
  - b) nozzle area ratio  $A_9/A_8$  for a perfectly expanded nozzle
  - c) nozzle exit Mach number  $M_9$  (perfectly expanded)

# Performance Evaluation of a Turbojet Engine

- As outlined in the earlier lecture, the performance parameters for an aircraft engine are
  - specific thrust
  - thrust specific fuel consumption (TSFC)
  - thermal, propulsive, and overall efficiencies.



# Aircraft Engine Performance

Cycle analysis



# Cycle Analysis

The approach to calculating the performance parameters of an aircraft engine is to march through an engine, component by component, until we calculate the target parameters, which are

- a) fuel-to-air ratio  $f$
- b) exhaust velocity  $V_9$ .



Flight Condition: Specifics of Mach number (or speed), static temperature, and pressure at the flight altitude.



Gas behaviour: Specific heats, gas constants, fuel heating value, etc

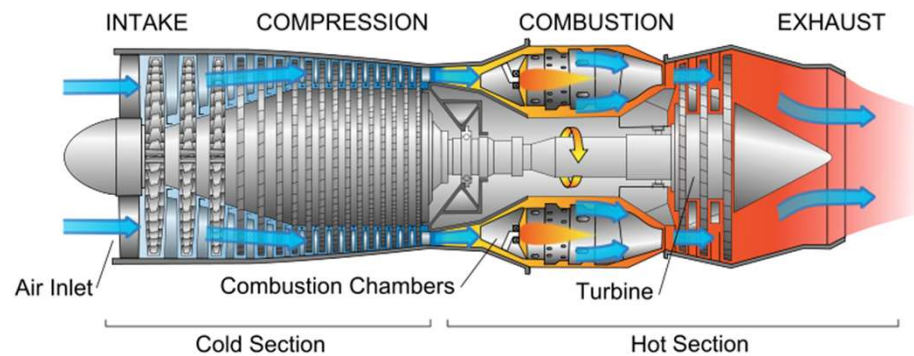


Design Characteristics/Constraints: Specifics of the engine technology that will be analyzed.

Example:

The following data correspond to a two spool engine turbojet. Calculate the specific thrust and specific fuel consumption to fit the gas generator and a convergent nozzle for the following conditions.

Diagram



## Characteristics

|                       |        |         |
|-----------------------|--------|---------|
| Flight conditions     |        |         |
| Static                |        |         |
| Temperature           | C      | 15      |
| Pressure              | kPa    | 101.325 |
| M0                    | -      | 0       |
| Gas behaviour         |        |         |
| Cp Cold Section       | J/kg.K | 1004    |
| $\gamma$ cold         | -      | 1.4     |
| Cp Hot Section        | J/kg.K | 1156    |
| $\gamma$ hot          | -      | 1.33    |
| FHV                   | kJ/kg  | 42.000  |
| Design                |        |         |
| Pressure Ratios       |        |         |
| Inlet                 | -      | 0.985   |
| Compressor            | -      | 15      |
| Burner                | -      | 0.95    |
| Nozzle                | -      | 0.98    |
| Efficiencies          |        |         |
| Compressor Isentropic | -      | 0.85    |
| Burner                | -      | 0.98    |
| Turbine Isentropic    | -      | 0.85    |
| Shaft/Mechanical      | -      | 0.995   |
| TIT                   | K      | 1480    |
| Air mass flow rate    | kg/s   | 150     |

# Inlet

First, the analysis for this engine is for uninstalled thrust conditions, nonetheless an estimated inlet pressure recovery data is given to establish inlet performance.

Therefore, the inlet discharge pressure must be calculated, this will be the pressure times the pressure ratio ( $P_{\text{exit}}/P_{\text{inlet}}$ ).

$$T_{t0} = 288.15 \text{ K}$$

$$P_{t0} = 101.33 \text{ kPa}$$

$$a_0 = 340.18 \text{ m/s} ; V_0 = 0 \text{ m/s};$$

$$P_{t2} = P_{t0} \pi_d = 101.33 \text{ kPa} \cdot 0.985 = 99.81 \text{ kPa}$$

# Low Pressure Compressor

With no inlet data, we can safely assume ambient conditions equal compressor inlet conditions

$$T_{t2} = 288.15 \text{ K}$$

$$P_{t2} = 99.81 \text{ kPa}$$

The compressor discharge pressure is the pressure times the pressure ratio ( $P_{\text{exit}}/P_{\text{inlet}}$ ).

$$P_{t2.5} = P_{t2} \pi = 99.81 \text{ kPa} \cdot 3.873 = 386.54 \text{ kPa}$$

From isentropic relations we can find the compressor discharge temperature in ideal conditions.

$$T_{t2.5 \text{ ideal}} = T_{t2} \left( \frac{P_{t2.5}}{P_{t2}} \right)^{\frac{\gamma-1}{\gamma}} = 424.26 \text{ K}$$

The deviation of the actual behavior of the compressor and the turbine from the ideal is considered by the isentropic efficiency, where:

$$\eta_{\text{is}} = \frac{T_{t2.5 \text{ ideal}} - T_{t2}}{T_{t2.5} - T_{t2}} \left( \frac{\text{Ideal}}{\text{Real}} \right) \Rightarrow T_{t2.5} = T_{t2} + \frac{T_{t2.5 \text{ ideal}} - T_{t2}}{\eta_{\text{is}}} = 288 + \frac{424.03 - 288}{0.85} = 448.28 \text{ K}$$

$$W_{\text{C}_{\text{LP}}} = \dot{m}_0 C_{p_c} (T_{t2.5} - T_{t2}) = 24.115 \text{ MW}$$

Shaft power consumed by the LPC

# High Pressure Compressor

$$P_{t3} = P_{t2.5} \pi = 1497.08 \text{ kPa}$$

Discharge pressure from LP compressor becomes the input for the HP compressor. As in previous step the pressure will be multiplied by the pressure ratio and the temperature calculated from isentropic relations.

$$T_{t3 \text{ ideal}} = T_{t2.5} \left( \frac{P_{t3}}{P_{t2.5}} \right)^{\frac{\gamma-1}{\gamma}} = 660.03 \text{ K}$$

$$\eta_{is} = \frac{T_{t3 \text{ ideal}} - T_{t2.5}}{T_{t3} - T_{t2.5}} \Rightarrow T_3 = T_{t2.5} + \frac{T_{t3 \text{ ideal}} - T_{t2.5}}{\eta_{is}} = 697.39 \text{ K}$$

$$W_{C_{HP}} = \dot{m}_0 C_{p_c} (T_{t3} - T_{t2.5}) = 37.516 \text{ MW}$$

HP compressor shaft power

# Combustor

$$\eta_{cc} = \frac{(\dot{m}_0 + \dot{m}_{fuel})h_4 - \dot{m}_0h_3}{\dot{m}_{fuel}(LHV)} \left( \frac{\Delta h_{real}}{\Delta h_{ideal}} \right)$$

$$\Rightarrow \dot{m}_{fuel} = \frac{\dot{m}_{air}(h_4 - h_3)}{(\eta_{cc} \cdot LHV - h_4)} = 3.843 \frac{\text{kg}}{\text{s}}$$

Combustor efficiency is a given value. Therefore the fuel mass flow can be calculated.

$$Q_{cc} = \Delta h_{real} = (\dot{m}_0 + \dot{m}_{fuel})h_4 - \dot{m}_0h_3 = 158.18 \text{ MW}$$

The same way the heat input is calculated as follows

# High Pressure Turbine

The power absorbed by the compressor is equivalent to the one generated in the turbine taking into consideration the mechanical losses

$$W_{\text{CHP}} = \eta_M \times W_{\text{THP}} = \eta_M \times \dot{m}_H c_{pH} (T_{t4} - T_{t5})$$

The total mass flow going through the turbine is the result of the air mass flow and the fuel flow.

$$\dot{m}_9 = \dot{m}_0 + \dot{m}_{\text{fuel}}$$

Therefore, the turbine exit temperature is calculated from the work.

$$T_{t4.5} = T_{t4} - \frac{W_{\text{CHP}}}{\eta_M \dot{m}_9 c_{pH}} = 1267.99 \text{ K}$$

If the loss in the pressure is equivalent to 5%, as in the compressor using the isentropic relations, the expansion can be calculated.

$$P_{t4} = 0.95 P_{t3} = 1422.22 \text{ kPa}$$

$$\eta_{\text{is}} = \frac{T_{t4} - T_{t4.5}}{T_{t4} - T_{t4.5 \text{ ideal}}} \left( \frac{\text{Real}}{\text{Ideal}} \right) \Rightarrow T_{t5 \text{ ideal}} = T_{t4} - \frac{T_{t4} - T_{t4.5}}{\eta_{\text{is}}} = 1230.57 \text{ K}$$

$$\frac{T_{t4.5 \text{ ideal}}}{T_{t4}} = \left( \frac{P_{t4.5}}{P_{t4}} \right)^{\frac{\gamma-1}{\gamma}} \Rightarrow P_{t4.5} = P_{t4} \left( \frac{T_{t4.5 \text{ ideal}}}{T_{t4}} \right)^{\frac{\gamma}{\gamma-1}} = 675.96 \text{ kPa}$$

# Low Pressure Turbine

$$W_{CLP} = \eta_M \times W_{ELP} = \eta_M \times \dot{m}_H C_{pH} (T_{t4.5} - T_{t5})$$

The process of the low-pressure turbine is completely analog to the HP turbine as follows.

$$T_{t5} = T_{t4.5} - \frac{W_{CHP}}{\eta_M \dot{m}_H C_{pH}} = 1131.70 \text{ K}$$

$$\eta_{is} = \frac{T_{t4.5} - T_{t5}}{T_{t4.5} - T_{t5 \text{ ideal}}} \Rightarrow T_{t5 \text{ ideal}} = T_{t4.5} - \frac{T_{t4.5} - T_{t5}}{\eta_{is}} = 1107.65 \text{ K}$$

$$P_{t5} = P_{t4.5} \left( \frac{T_{t5 \text{ ideal}}}{T_{t4.5}} \right)^{\frac{\gamma}{\gamma-1}} = 392.01 \text{ kPa}$$



# Engine Nozzle

Since the flow in the nozzle is adiabatic and there is no afterburner or additional heat input

$$T_{t9} = T_{t7} = T_{t5}$$
$$P_{t9} = \pi_n \cdot P_{t7} = 384.17 \text{ kPa}$$

$$NPR_{crit} = \left( \frac{P_{t9}}{P_0} \right)^* = \left[ \frac{\gamma + 1}{2} \right]^{\frac{\gamma}{\gamma - 1}} = 1.851$$

$$NPR_{real} = \left( \frac{P_{t9}}{P_0} \right) = 3.791$$

$NPR_{real} > NPR_{crit}$  nozzle is choked

$$P_9 = \frac{P_{t9}}{\left( \frac{P_{t9}}{P_{t0}} \right)^*} = 207.59 \text{ kPa}$$

where,

$P_9$  = static pressure

$$T_9 = \frac{T_{t9}}{\left( \frac{T_{t9}}{T_{t0}} \right)^*} = 971.42 \text{ K}$$

$T_9$  = static temperature, calculated with  $M=1$

# Thrust Specific Fuel Consumption

From static conditions in the nozzle, the speed of sound is calculated

$$\vec{V}_{\text{jet}} = \vec{V}_9 = \sqrt{\gamma_H R_{\text{aire}} T_9} = 567.3 \text{ m/s}$$

Density is calculated from the pressure and temperature of the exhaust gases

$$\rho_9 = \frac{P_9}{T_9 R_9} = 0.745 \text{ kg/m}^3$$

Therefore, area is as follows

$$A_9 = \frac{\dot{m}_H}{\rho_9 \vec{V}_9} = 0.363 \text{ m}^2$$

Thrust is calculated as:

$$FN = \dot{m}_H (\vec{V}_9 - V_o) + A_9(p_9 - p_o) = 125.956 \text{ kN}$$

Specific fuel consumption

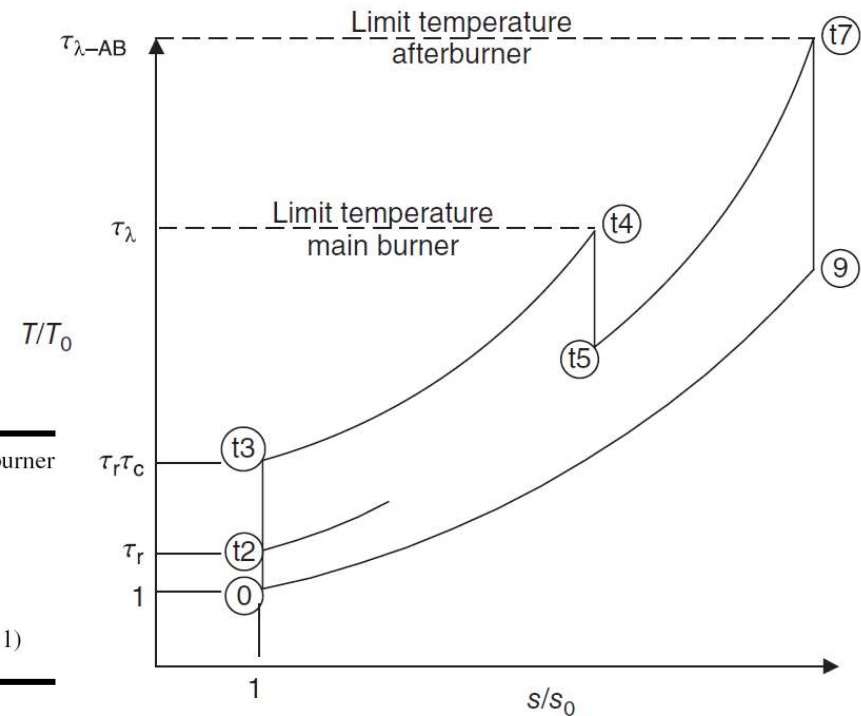
$$SFC = \frac{\dot{m}_f}{FN} = 109.84 \frac{\text{kg/h}}{\text{kN}}$$

# Performance Evaluation of a Turbojet Engine with Afterburner

The thrust of any given engine may be increased by raising the temperature of the exhaust gases. Fortunately, this is easy to do, since less than one-third of the total oxygen in the air flowing through the engine is consumed in the main combustion chamber, and the remainder is available in the turbine efflux gases to support further combustion. Thus, in order to obtain more thrust, it is only necessary to inject additional fuel into these gases to mix and burn with the available oxygen.

## New parameters in an afterburner

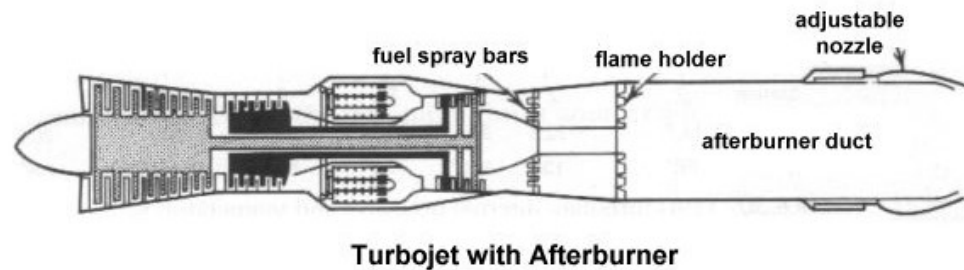
1.  $Q_{R,AB}$  (ideal) heating value of the fuel in the afterburner
2.  $\dot{m}_{f,AB}$  or  $T_{t7}$  or  $\tau_{\lambda-AB} \equiv \frac{c_{pAB} T_{t7}}{c_{pc} T_0}$  fuel flow rate or the exit temperature
3.  $\eta_{AB} \equiv \frac{(Q_{R,AB})_{actual}}{(Q_{R,AB})_{ideal}}$  afterburner efficiency ( $<1$ )
4.  $\pi_{AB} \equiv \frac{p_{t7}}{p_{t5}}$  total pressure ratio across the afterburner ( $<1$ )



Example 4.11:

The following data correspond to a single spool engine turbojet with afterburner. Calculate the specific thrust, specific fuel consumption, thermal efficiency, propulsive efficiency and overall efficiency to fit the gas generator and a perfectly expanded convergent nozzle for the following conditions.

Diagram



What are the main differences between this and the prior example?

## Characteristics

|                       |        |        |
|-----------------------|--------|--------|
| Flight conditions     |        |        |
| Dynamic               |        |        |
| Temperature           | C      | -45    |
| Pressure              | kPa    | 10     |
| M0                    | -      | 2      |
| Gas behaviour         |        |        |
| Cp Cold Section       | J/kg.K | 1004   |
| $\gamma$ cold         | -      | 1.4    |
| Cp Hot Section        | J/kg.K | 1156   |
| $\gamma$ hot          | -      | 1.33   |
| Cp AB Section         | J/kg.K | 1243   |
| $\gamma$ AB           | -      | 1.30   |
| FHV                   | kJ/kg  | 42.000 |
| Design                |        |        |
| Pressure Ratios       |        |        |
| Diffuser              | -      | 0.88   |
| Compressor            | -      | 12     |
| Burner                | -      | 0.95   |
| Afterburner           | -      | 0.93   |
| Nozzle                | -      | 0.93   |
| Efficiencies          |        |        |
| Compressor Polytropic | -      | 0.9    |
| Burner                | -      | 0.98   |
| Turbine Polytropic    | -      | 0.82   |
| Afterburner           | -      | 0.98   |
| Shaft/Mechanical      | -      | 0.995  |
| TIT                   | K      | 1585   |
| AB TIT                | K      | 2025   |

# Inlet

First of all, the analysis for this engine is for installed thrust conditions, therefore the Total conditions of the gases need to be calculated

The inlet discharge pressure is the pressure times the pressure ratio ( $P_{\text{exit}}/P_{\text{inlet}}$ ).

$$T_{t0} = 410.7 \text{ K}$$

$$P_{t0} = 78.244 \text{ kPa}$$

$$a_0 = 302.6 \text{ m/s} ; V_0 = 605.2 \text{ m/s}$$

$$P_{t2} = P_{t0} \pi_d = 78.24 \text{ kPa} \cdot 0.88 = 68.855 \text{ kPa}$$

# Compressor

The compressor discharge pressure is the pressure times the pressure ratio ( $P_{\text{exit}}/P_{\text{inlet}}$ ).

$$P_{t3} = P_{t2} \pi = 68.85 \text{ kPa} \cdot 12 = 826.262 \text{ kPa}$$

For this example, polytopic efficiency is given instead of isentropic efficiency, therefore the compressor discharge temperature can be calculated as follows:

$$T_{t3} = T_{t2} \left( \frac{P_{t3}}{P_{t2}} \right)^{\frac{\gamma-1}{\gamma \cdot \eta_c}} = 903.8 \text{ K}$$

# Combustor

$$\eta_{cc} = \frac{(\dot{m}_{air} + \dot{m}_{fuel})h_4 - \dot{m}_{air}h_3}{\dot{m}_{fuel}(LHV)} \left( \frac{\Delta h_{real}}{\Delta h_{ideal}} \right)$$
$$\Rightarrow f = \frac{(h_4 - h_3)}{(\eta_{cc} \times LHV - h_4)} = 0.02352$$

If the loss in the pressure is equivalent to 5%, as in the compressor using the isentropic relations, the expansion can be calculated.

Combustor efficiency is a given value. And since the air massflow is unknown the equation can be solved for fuel to air ratio.

$$P_{t4} = 0.95P_{t3} = 784.949 \text{ kPa}$$

# Turbine

The power absorbed by the compressor is equivalent to the one generated in the turbine taking into consideration the mechanical losses, although since the air mass flow going into the engine is unknown, the power balance must be done using the fuel to air ratio

$$W_{\text{CHP}} = \eta_M \times W_{\text{EHP}} = \eta_M \times (1 + f) C_{pH} (T_{t4} - T_{t5})$$

Therefore, the turbine exit temperature is calculated from the work.

$$T_{t5} = T_{t4} - \frac{C_{pC} (T_{t3} - T_{t2})}{C_{pH} \eta_M (1 + f)} = 1164.42 \text{ K}$$

Using polytropic efficiency and the known turbine temperature ratio, we can calculate the turbine pressure ratio

$$\frac{T_{t5}}{T_{t4}} = \left( \frac{P_{4.5}}{P_4} \right)^{\frac{(\gamma-1) \cdot e_t}{\gamma}} \Rightarrow P_5 = P_4 \left( \frac{T_{t5}}{T_{t4}} \right)^{\frac{\gamma}{(\gamma-1) \cdot e_t}} = 172.43 \text{ kPa}$$



# Afterburner

$$\eta_{AB} = \frac{(\dot{m}_{air} + \dot{m}_{fuel} + \dot{m}_{AB})h_7 - (\dot{m}_{fuel} + \dot{m}_{air})h_5}{\dot{m}_{AB}(\text{LHV})} \left( \frac{\Delta h_{real}}{\Delta h_{ideal}} \right)$$

$$\Rightarrow f_{AB} = \frac{(1 + f)(h_7 - h_5)}{(\eta_{ab} \times \text{LHV} - h_5)} = 0.03102$$

If the loss in the pressure is equivalent to 5%, as in the compressor using the isentropic relations, the expansion can be calculated.

As before, the afterburner is analog to the combustor analysis .

$$P_{t7} = 0.93P_{t5} = 160.36 \text{ kPa}$$

# Nozzle

The first thing to analyze is the nozzle pressure recovery

$$P_{t9} = 0.93 P_{t7} = 0.93 \cdot 160.36 = 149.13 \text{ kPa}$$

It is established in the problem that the nozzle is convergent and perfectly expanded, meaning that  $p_9 = p_0$ . Therefore the rest of the properties can be calculated based on such assumption.

$$M_9 = \sqrt{\frac{2}{\gamma_{AB} - 1} \left[ \left( \frac{P_9}{p_9} \right)^{\frac{\gamma_{AB}-1}{\gamma_{AB}}} - 1 \right]} = 2.40$$

Static Temperature

$$T_9 = \frac{T_{t9}}{1 + \left( \frac{\gamma_{AB} - 1}{2} M_9^2 \right)}$$

Speed of Sound

$$a_9 = \sqrt{(\gamma_{AB} - 1) C p_{AB} T_9} = 636.2 \text{ m/s}$$

Jet speed

$$V_9 = M_9 \cdot a_9 = 1528.3 \text{ m/s}$$

# Performance Parameters

Specific Thrust

$$\frac{F_n}{\dot{m}_0} = (1 + f + f_{AB})V_9 - V_0 = 1006.28 \frac{N}{kg/s}$$

Thrust Specific Fuel Consumption

$$TSFC = \frac{f + f_{AB}}{F_n/\dot{m}_0} = 54.191 \frac{mg}{N}$$

Thermal Efficiency

$$\eta_{TH} = \frac{(1 + f + f_{AB}) \frac{V_9^2}{2} - \frac{V_0^2}{2}}{f \cdot Q_R + f_{AB} \cdot Q_{RAB}} = 45.77 \%$$

Propulsive Efficiency

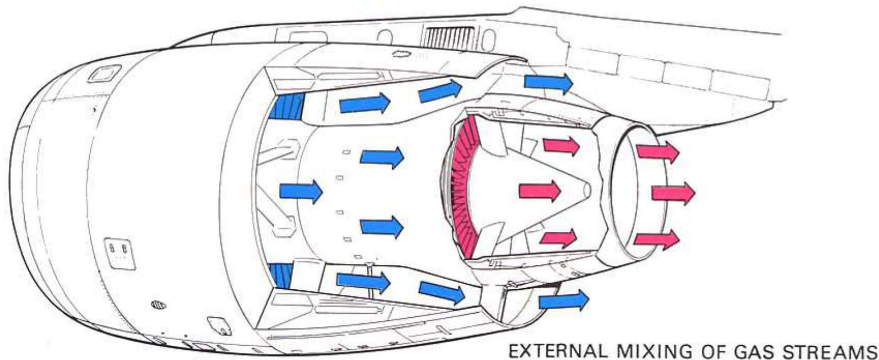
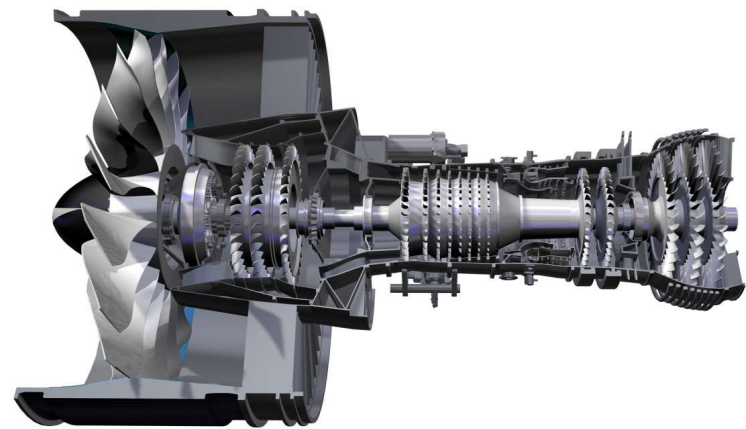
$$\eta_p = \frac{(F_n/\dot{m}_0)/V_0}{(1 + f + f_{AB}) \frac{V_9^2}{2} - \frac{V_0^2}{2}} = 58.11 \%$$

Overall Efficiency

$$\eta_{ov} = \eta_{th} \cdot \eta_p = 26.60 \%$$

# Performance Evaluation of a Separate – Exhaust Turbofan Engine

- To create a turbofan engine, a basic gas generator is followed by an additional turbine stage(s), which tap into the exhaust stream thermal energy to provide shaft power to a fan.



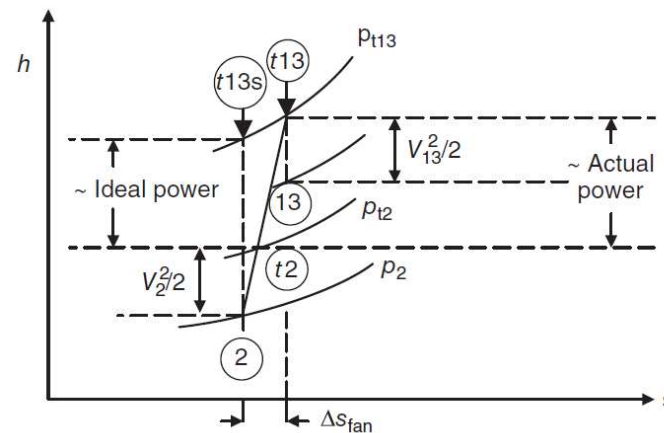
- Cold by-pass (fan) airflow
- Hot exhaust gases

# Analysis of a Separate Exhaust Turbofan Engine

- How do we treat the thermodynamics of a fan?
  - Answer: As a compressor, the fan is characterized by its pressure ratio and efficiency (isentropic or polytropic). Therefore, the same set of equations shall apply.

Inlet adiabatic efficiency

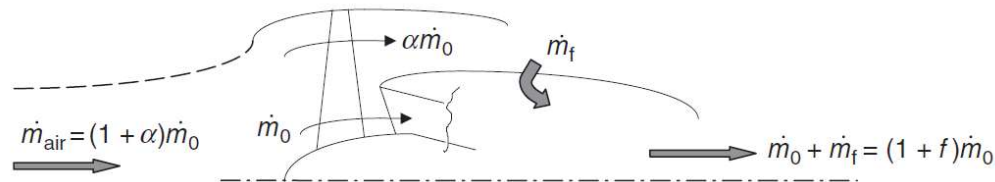
$$\eta_c \equiv \frac{h_{t13s} - h_2}{h_{t13} - h_2} = \frac{\Delta h_{t,isentropic}}{\Delta h_{t,actual}}$$



# Energy Balance of a Separate Exhaust Turbofan Engine

- Energy Balance
  - The power balance of a turbofan depends entirely on the shaft configuration, whether it's a multi spool or a single spool engine, we need to understand the power source of the fan.
    - Note that the mass flow rates in a turbofan engine are
      - $\dot{m}_0$  air mass flow rate through the engine core
      - $\alpha\dot{m}_0$  air mass flow rate through the fan duct
      - $(1 + \alpha)\dot{m}_0$  air mass flow rate through the inlet (which later splits into core and the fan duct flow rates)
      - $\dot{m}_f$  the fuel flow rate in the main burner and the fuel-to-air ratio  $f$
  - The energy balance therefore is

$$\eta_M \dot{m}_0 (1 + f) (h_{t4} - h_{t5}) = \dot{m}_0 (h_{t3} - h_{t2}) + \alpha \dot{m}_0 (h_{t13} - h_{t2})$$



# Thrust in a Separate Exhaust Turbofan

- By applying the same momentum principles as in the turbojet but for two separate streams and accounting the gross thrust for two nozzles and account for the entire ram through the inlet the following equation is obtained:

$$F_n = \underbrace{\dot{m}_0(1+f)V_9 + \alpha\dot{m}_0V_{19}}_{\text{Momentum Change}} + \underbrace{(p_9 - p_0)A_9 + (p_{19} - p_0)A_{19}}_{\text{Flow Expansion}} - \underbrace{(1+\alpha)\dot{m}_0V_0}_{\text{Ram Drag}}$$

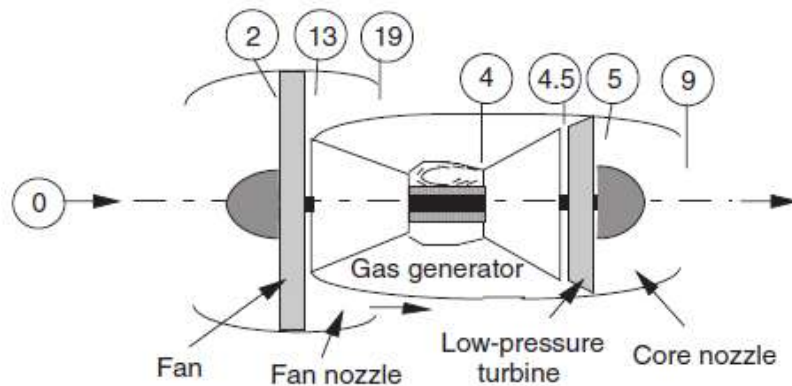
- In case that the exhaust nozzles are not perfectly expanded, we may use the effective exhaust speeds in place of actual exit velocities which are defined as follows:

$$V_{19\text{eff}} = \frac{F_{g-fan}}{\dot{m}_{19}}; V_{9\text{eff}} = \frac{F_{g-fan}}{\dot{m}_9}$$

Example 4.13:

The following data correspond to a single spool engine turbofan with separate exhaust. Calculate the total pressure and temperatures throughout the engine. Nozzle exit static pressures, exit velocities, non dimensional specific thrust, TSFC and all engine efficiencies. Nozzle are both convergent.

Diagram



### Characteristics

|                   |    |       |
|-------------------|----|-------|
| Flight conditions |    |       |
| Dynamic           |    |       |
| Temperature       | C  | -40   |
| Pressure          | Pa | 15000 |
| M0                | -  | 0.88  |

|                 |        |       |
|-----------------|--------|-------|
| Gas Behaviour   |        |       |
| Cp Cold Section | J/kg.K | 1004  |
| $\gamma$ cold   | -      | 1.4   |
| Cp Hot Section  | J/kg.K | 1152  |
| $\gamma$ hot    | -      | 1.33  |
| FHV             | kJ/kg  | 42800 |

### Design

|                              |   |          |       |
|------------------------------|---|----------|-------|
| Pressure Ratios              |   | Core     | Fan   |
| Diffuser                     | - | 0.995    | 0.995 |
| Compressor                   | - | 40       | 1.6   |
| Burner                       | - | 0.95     |       |
| Nozzle                       | - | 0.98     | 0.98  |
| Efficiencies                 |   |          |       |
| Compressor Polytropic        | - | 0.9      | 0.9   |
| Burner                       | - | 0.992    |       |
| Turbine Polytropic (HP & LP) | - | 0.85     |       |
| Shaft/Mechanical (HP & LP)   | - | 0.95     |       |
| TIT                          | K | 1624.528 |       |
| Bypass Ratio                 |   | 8        |       |



# Inlet

As in the previous example for the afterburner, the analysis for this engine is for installed thrust conditions, therefore the total conditions of the gases need to be calculated

The inlet discharge pressure is the pressure times the pressure ratio ( $P_{\text{exit}}/P_{\text{inlet}}$ ).

$$T_{t0} = 269.26 \text{ K}$$

$$P_{t0} = 24.83 \text{ kPa}$$

$$a_0 = 305.99 \text{ m/s} ; V_0 = 269.27 \text{ m/s};$$

$$P_{t2} = P_{t0} \pi_d = 24.83 \text{ kPa} \cdot 0.995 = 24.70 \text{ kPa}$$

# Fan

The fan works as another compressor; therefore, the discharge pressure is the pressure times the fan pressure ratio ( $P_{\text{fan\_exit}}/P_{\text{inlet}}$ ).

$$P_{t13} = P_{t2} \pi = 24.70 \text{ kPa} \cdot 1.6 = 39.53 \text{ kPa}$$

For this example, polytropic efficiency is given instead of isentropic efficiency, therefore the compressor discharge temperature can be calculated as follows:

$$T_{t13} = T_{t2} \left( \frac{P_{t13}}{P_{t2}} \right)^{\frac{\gamma-1}{\gamma \cdot e_c}} = 312.58 \text{ K}$$

# Compressor

The compressor discharge pressure is the pressure times the pressure ratio ( $P_{\text{exit}}/P_{\text{inlet}}$ ).

$$P_{t3} = P_{t2} \pi = 24.71 \text{ kPa} \cdot 40 = 98.82 \text{ kPa}$$

For this example, polytopic efficiency is given instead of isentropic efficiency, therefore the compressor discharge temperature can be calculated as follows:

$$T_{t3} = T_{t2} \left( \frac{P_{t3}}{P_{t2}} \right)^{\frac{\gamma-1}{\gamma \cdot \eta_c}} = 868.48 \text{ K}$$

# Combustor

$$\eta_{cc} = \frac{(\dot{m}_{air} + \dot{m}_{fuel})h_4 - \dot{m}_{air}h_3}{\dot{m}_{fuel}(LHV)} \left( \frac{\Delta h_{real}}{\Delta h_{ideal}} \right)$$
$$\Rightarrow f = \frac{(h_4 - h_3)}{(\eta_{cc} \times LHV - h_4)} = 0.02463$$

If the loss in the pressure is equivalent to 5%, as in the compressor using the isentropic relations, the expansion can be calculated.

Combustor efficiency is a given value. And since the air mass flow is unknown the equation can be solved for fuel to air ratio.

$$P_{t4} = 0.95P_{t3} = 93.88 \text{ kPa}$$

# Turbine

The power absorbed by the compressor is equivalent to the one generated in the turbine taking into consideration the mechanical losses, although since the air mass flow going into the engine is unknown, the power balance must be done using the fuel to air ratio

$$W_{\text{CHP}} + W_{\text{FHP}} = \eta_M \times W_{\text{EHP}} = \eta_M \times (1 + f) C_{pH} (T_{t4} - T_{t5})$$

$$T_{t5} = T_{t4} - \frac{C_{p_c} (T_{t3} - T_{t2}) + (1 + \alpha) C_{p_c} (T_{13} - T_{12})}{C_{pH} \eta_M (1 + f)}$$

Therefore the turbine exit temperature is calculated from the work.

$$T_{t5} = 777.66 \text{ K}$$

Using polytropic efficiency and the known turbine temperature ratio, we can calculate the turbine pressure ratio

$$\frac{T_{t5}}{T_{t4}} = \left( \frac{P_{4.5}}{P_4} \right)^{\frac{(\gamma-1) \cdot e_t}{\gamma}} \Rightarrow P_5 = P_4 \left( \frac{T_{t5}}{T_{t4}} \right)^{\frac{\gamma}{(\gamma-1) \cdot e_t}} = 28.55 \text{ kPa}$$

# Core Nozzle

The first thing to analyze is the nozzle pressure recovery

$$P_{t9} = 0.98 P_{t5} = 0.98 \cdot 28.55 \text{ kPa} = 27.98 \text{ kPa}$$

Considering the nozzle to be convergent, maximum Mach number it can be achieved is  $M_9 = 1$ .

$$T_{t9} = T_{t5} = 312.58 \text{ K}$$

Since the nozzle is considered to be adiabatic, the effect of heat transfer is neglected meaning

$$p_9 = \frac{p_{t9}}{\left[1 + \frac{\gamma_c - 1}{2}\right]^{\frac{\gamma_c}{\gamma_c - 1}}} = 19.838 \text{ kPa}$$

As stated earlier the use of exhaust effective velocity is suggested for nozzles that are not perfectly expanded

$$V_{9\text{eff}} = V_9 + \frac{(p_9 - p_0)A_9}{\dot{m}_9} = 379.78 \text{ m/s}$$

# Fan Nozzle

The same procedure as for the core nozzle is followed

First the nozzle pressure recovery is taken into account to determine the total exit pressure of the fan

$$P_{t19} = P_{t13} \pi_{fn} = 39.528 \text{ kPa} \cdot 0.95 = 37.552 \text{ kPa}$$

Since the nozzle is considered to be adiabatic, the effect of heat transfer is neglected meaning

$$T_{t19} = T_{t1} = 312.58 \text{ K}$$

Since the nozzle is convergent, the highest exit Mach number that it can attain is Mach 1. By calculating the static pressure it can be verified whether the nozzle is choked and underexpanded.

$$p_{19} = \frac{p_{t19}}{\left[1 + \frac{\gamma_c - 1}{2}\right]^{\frac{\gamma_c}{\gamma_c - 1}}} = 19.838 \text{ kPa} \quad p_{19} > p_0$$

As stated earlier the use of exhaust effective velocity is suggested for nozzles that are not perfectly expanded;

$$V_{19\text{eff}} = V_{19} + \frac{(p_{19} - p_0)A_{19}}{\dot{m}_{19}} = 379.78 \text{ m/s}$$

# Performance Parameters

Since the effective velocity was previously calculated, they both can be used to calculate the Specific Thrust (non-dimensional).

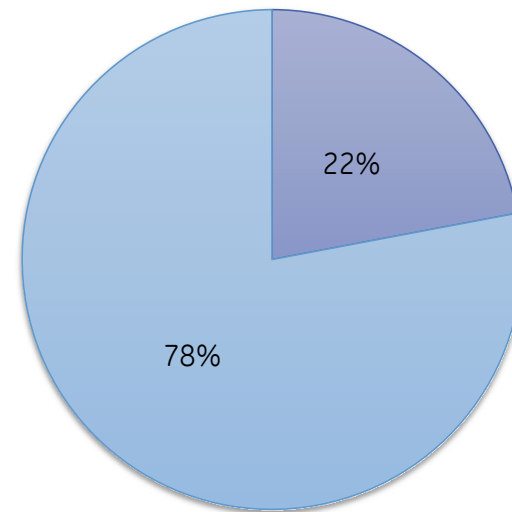
Specific Thrust – Fan

$$\frac{F_{n-co}}{(1 + \alpha)\dot{m}_0 \cdot a_0} = 0.091$$

$$\frac{F_{n-fan}}{(1 + \alpha)\dot{m}_0 \cdot a_0} = 0.321$$

Specific Thrust – Total

$$\frac{F_{n-cor} + F_{n-fa}}{(1 + \alpha)\dot{m}_0 \cdot a_0} = 0.412$$



■ % Core Thrust Contribution ■ % Fan Thrust Contribution



# Performance Parameters

Thrust Specific Fuel Consumption

$$TSFC = \frac{f}{(1+\alpha)a_0 F_{n-total} / (1+\alpha)\dot{m}_0 \cdot a_0} = 21.72$$

Thermal Efficiency

$$\eta_{TH} = \frac{\Delta KE}{\dot{m}_f Q_R} = 36.26 \%$$

Propulsive Efficiency

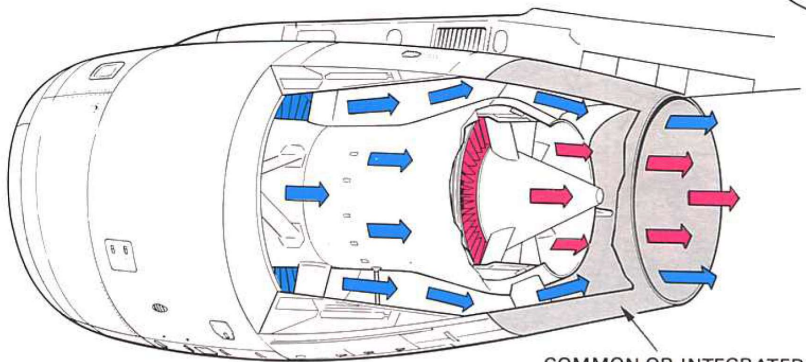
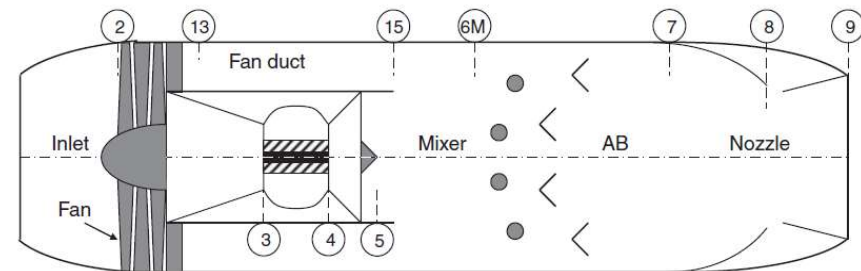
$$\eta_p = \frac{F \cdot V_0}{\Delta KE} = 79.89 \%$$

Overall Efficiency

$$\eta_{ov} = \eta_{th} \cdot \eta_p = 28.97 \%$$

# Performance Evaluation of a Mixed-Exhaust Turbofan Engine with an Afterburner

- The engine core discharges a hot gas into the mixer, whereas the fan duct injects a cold gas into the mixer. Upon mixing of the two streams, the gas will attain a "mixed - out" state at station 6M.



■ Cold by-pass (fan) airflow  
■ Hot exhaust gases

PARTIAL INTERNAL MIXING OF GAS STREAMS

# Fan duct aerodynamics

- Kutta trailing – edge condition
  - A body with a sharp trailing edge which is moving through a fluid will create about itself a circulation of sufficient strength to hold the rear stagnation point at the trailing edge.
- How it applies?
  - Suggest that the two merging streams at the trailing edge of an airfoil, or fan duct and engine core streams can't support a static pressure jump. Therefore,
$$p_{15} = p_5$$
- Basically, the static pressure that is communicated between the core and the fan duct are set by the engine backpressure, and therefore the fan pressure ratio (hence the mass flow rate).
  - In other words, the two parameters of a fan, the pressure ratio and the bypass ratio can not be set independently in this configuration.

# Mixer

To analyze a constant – area mixer, we employ the conservation principles of mass, momentum and energy. Also, the mixture gas laws establish the mixed-out gas properties at the mixer exit.

- Mass Balance

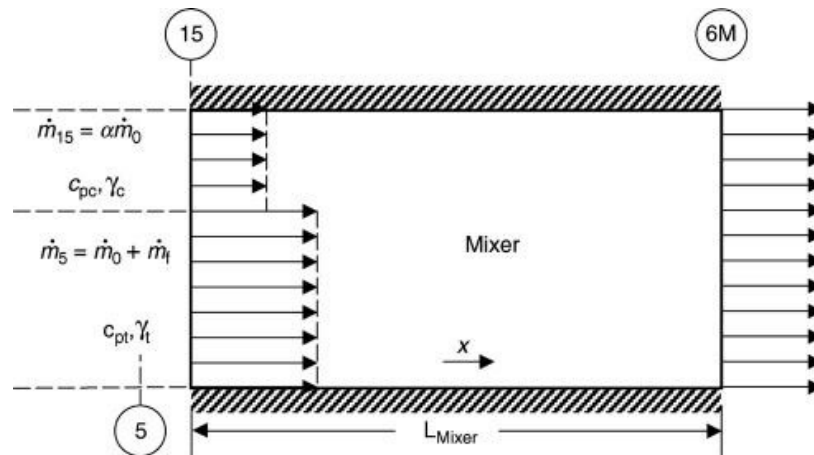
$$\dot{m}_5 + \dot{m}_{15} = \dot{m}_{6M}$$

- Momentum Balance\*

$$(\dot{m}_{6M} V_{6M} + p_{6M} A_{6M}) - (\dot{m}_5 V_5 + p_5 A_5) - (\dot{m}_{15} V_{15} + p_{15} A_{15}) = -D_{f_{mixer}}$$

- Energy Balance

$$\dot{m}_5 h_{t5} + \dot{m}_{15} h_{t15} = \dot{m}_{6M} h_{t6M}$$



$$\dot{m}_{6M} = (1 + \alpha) \dot{m}_0 + \dot{m}_t$$

Mixed-out gas properties

$$c_{p6M} = \frac{\dot{m}_5 c_{pt} + \dot{m}_{15} c_{pc}}{\dot{m}_5 + \dot{m}_{15}}$$

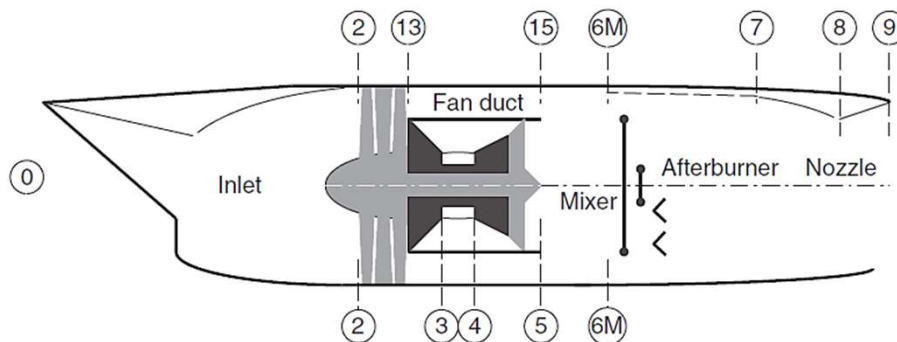
$$\gamma_{6M} = \frac{\dot{m}_5 c_{pt} + \dot{m}_{15} c_{pc}}{\dot{m}_5 \left( \frac{c_{pt}}{\gamma_t} \right) + \dot{m}_{15} \left( \frac{c_{pc}}{\gamma_c} \right)}$$

\* We note that the fluid momentum out of the mixer is somewhat less than the (sum of the) momentum entering the mixer and this difference or loss is caused by the friction drag on the mixer wall. Therefore, if a frictionless Flow in the mixer is assumed the momentum is conserved.

#### Example 4.16:

The following data correspond to a single spool engine turbofan with mixed exhaust. Calculate the total pressure and temperatures throughout the engine. Nozzle exit static pressures, exit velocities, non dimensional specific thrust, TSFC and all engine efficiencies.

Diagram



#### Characteristics

|                   |        |        |
|-------------------|--------|--------|
| Flight conditions |        |        |
| Dynamic           |        |        |
| Temperature       | C      | -50    |
| Pressure          | kPa    | 10     |
| M0                | -      | 2      |
| Gas behaviour     |        |        |
| Cp Cold Section   | J/kg.K | 1004   |
| $\gamma$ cold     | -      | 1.4    |
| Cp Hot Section    | J/kg.K | 1152   |
| $\gamma$ hot      | -      | 1.33   |
| Cp AB Section     | J/kg.K | 1241   |
| $\gamma$ AB       | -      | 1.30   |
| FHV               | kJ/kg  | 42.000 |

#### Design

##### Pressure Ratios

|                |   |        |
|----------------|---|--------|
| Diffuser       | - | 0.9    |
| Fan/Compressor | - | 1.9/13 |
| Burner         | - | 0.95   |
| Fan Duct       | - | 0.99   |
| Mixer          | - | 0.98   |
| Afterburner    | - | 0.93   |
| Nozzle         | - | 0.93   |
| Nozzle p9/p0   | - | 3.8    |

##### Efficiencies

|                           |   |      |
|---------------------------|---|------|
| Fan/Compressor Polytropic | - | 0.9  |
| Burner                    | - | 0.98 |
| Turbine Polytropic        |   | 0.80 |
| Afterburner               |   | 0.98 |
| Shaft/Mechanical          |   | 0.95 |
| TIT                       | K | 1585 |
| AB TIT                    | K | 2025 |
| Turbine exit Mach No      |   | 0.5  |

# Inlet

First, the analysis for this engine is for installed thrust conditions, therefore the total conditions of the gases need to be calculated

The inlet discharge pressure is the pressure times the pressure ratio ( $P_{\text{exit}}/P_{\text{inlet}}$ ).

$$T_{t0} = 401.7 \text{ K}$$

$$P_{t0} = 78.244 \text{ kPa}$$

$$a_0 = 299.4 \text{ m/s} ; V_0 = 598.7 \text{ m/s}$$

$$P_{t2} = P_{t0} \pi_d = 78.24 \text{ kPa} \cdot 0.90 = 70.420 \text{ kPa}$$

Remember inlet is treated as adiabatic and thus  $T_{t2} = T_{t0}$

# Fan

Fan exit pressure is given by the pressure ratio and fan exit temperature is related to it by the polytropic efficiency.

$$P_{t13} = P_{t2} \pi = 70.42 \text{ kPa} \cdot 1.9 = 133.798 \text{ kPa}$$

The fan duct is also adiabatic this means  $T_{t13} = T_{t15}$

$$T_{t13} = T_{t2} \left( \frac{P_{t13}}{P_{t2}} \right)^{\frac{\gamma-1}{\gamma \cdot e_c}} = 492.45 \text{ K}$$

# Compressor

The compressor calculations are like the previous examples

$$P_{t3} = P_{t2} \pi = 70.42 \text{ kPa} \cdot 13 = 915.46 \text{ kPa}$$

$$T_{t3} = T_{t2} \left( \frac{P_{t3}}{P_{t2}} \right)^{\frac{\gamma-1}{\gamma \cdot e_c}} = 906.78 \text{ K}$$



# Combustor

$$\eta_{cc} = \frac{(\dot{m}_{air} + \dot{m}_{fuel})h_4 - \dot{m}_{air}h_3}{\dot{m}_{fuel}(LHV)} \left( \frac{\Delta h_{real}}{\Delta h_{ideal}} \right)$$
$$\Rightarrow f = \frac{(h_4 - h_3)}{(\eta_{cc} \times LHV - h_4)} = 0.02373$$

If the loss in the pressure is equivalent to 5%, as in the compressor using the isentropic relations, the expansion can be calculated.

Combustor efficiency is a given value. And since the air mass flow is unknown the equation can be solved for fuel to air ratio.

$$P_{t4} = 0.95P_{t3} = 869.687 \text{ kPa}$$

# Turbine/Fan Duct Exit

Unlike the previous example, the energy balance for the turbine will be imposed by the exit pressure, which needs to be balanced with the fan duct exit.

$$P_{t15} = P_{t5}$$

Therefore, the turbine exit pressure is calculated from the different pressures.

$$\frac{P_{t5}}{P_{t4}} = \frac{\pi_{FD} \cdot \pi_f}{\pi_b \cdot \pi_c} = 0.1523 \text{ kPa}$$

Using polytropic efficiency and the known turbine pressure ratio, we can calculate the turbine pressure ratio

$$\frac{T_{t5}}{T_{t4}} = \left( \frac{P_{t5}}{P_{t4}} \right)^{\frac{(\gamma-1) \cdot e_t}{\gamma}} \Rightarrow T_{t5} = T_{t4} \left( \frac{P_{t5}}{P_{t4}} \right)^{\frac{(\gamma-1) \cdot e_t}{\gamma}} = 1101.27 \text{ kPa}$$

# Turbine/Fan Duct Exit

Now, with the total temperature and pressure already known, the energy balance around the shaft power can be used to calculate the bypass ratio.

$$W_{\text{CHP}} + W_{\text{FHP}} = \eta_M \times W_{\text{EHP}}$$

$$(h_{t3} - h_{t2}) + \alpha(h_{t13} - h_{t12}) = \eta_M \times (1 + f)C_{pH}(T_{t4} - T_{t5})$$

Since engine backpressure dictates the mass flow rate and bypass ratio for the engine, the only missing parameter to calculate in this station is the duct fan Mach number.

$$\alpha = 0.5665$$

$$M_{15} = \left[ \frac{2}{\gamma_c - 1} \left\{ \left[ \left( 1 + \frac{\gamma_t - 1}{2} M_5^2 \right)^{\frac{\gamma_t}{\gamma_t - 1}} \right]^{\frac{\gamma_c - 1}{\gamma_c}} \right\} - 1 \right]^{1/2} = 0.488$$

# Mixer

The mixer is the crucial part in the analysis of this engine, a mass, momentum and energy balance must be carried out in this section before proceeding to the next one.

The energy balance will help finding the total enthalpy out of the mixer

$$\dot{m}_{6M} h_{t6M} = \dot{m}_5 h_{t5} + \dot{m}_{15} h_{t15}$$

$$h_{t6M} = \frac{(1+f)h_{t5} + \alpha h_{t15}}{1 + \alpha + f} = 992.857 \frac{kJ}{kg}$$

Same for the specific heats and the gamma

$$C_{p6M} = 1099.3 \frac{kJ}{kg K} \text{ and } \gamma_{6M} = 1.352$$

The gas temperature and static pressure at the fan duct and the turbine are calculated with the total properties and the Mach numbers

$$p_5 = p_{15} = 112.547 \text{ kPa}$$

$$T_5 = 1057.64 \text{ K and } T_{15} = 473.83 \text{ K}$$

The area ratio between the fan duct and the turbine exit flow is expressed via the continuity equation (mass flow balance):

$$\frac{A_{15}}{A_5} = \frac{\alpha}{1+f} \cdot \left( \frac{\gamma_t}{\gamma_c} \right) \frac{a_{15}}{a_5} \frac{M_5}{M_{15}} = 0.3604$$

# Mixer

The momentum balance is a bit of a more elaborate step, its needed in order to obtain the Mach number at the mixer, along with the mass balance

From momentum balance

$$\frac{p_{6M}}{p_5} [1 + \gamma_{6M} M_{6M}^2] = \frac{[(1 + \gamma_t M_5^2) + (A_{15}/A_5)(1 + \gamma_c M_{15}^2)]}{(1 + A_{15}/A_5)} = C1 = 1.3327$$

From mass balance and non dimensional mass flow equation

$$\frac{p_{6M}}{p_5} \left( M_{6M} \sqrt{1 + (\gamma_{6M} - 1) M_{6M}^2 / 2} \right) = \left[ \left( \frac{\gamma_t}{\gamma_{6M}} \right) \frac{M_5}{a_5} + \left( \frac{\gamma_c}{\gamma_{6M}} \right) \frac{M_{15} (A_{15}/A_5)}{a_{15}} \right] \frac{\sqrt{(\gamma_{6M} - 1) C_{p6M} T_{t6M}}}{(1 + A_{15}/A_5)} = C2 = 0.52359$$

By dividing both equations, the mixer static pressure ratio is eliminated and arrive at the mixer exit Mach number via

$$\frac{1 + \gamma_{6M} M_{6M}^2}{M_{6M} \sqrt{1 + (\gamma_{6M} - 1) M_{6M}^2 / 2}} = \frac{C_1}{C_2} = \sqrt{C}$$

# Mixer

The previous equation leaves a quadratic to be solved for  $M_{6M}$ , expressed as follows:

$$M_{6M}^2 = \frac{C - 2\gamma_{6M} - \sqrt{(C - 2\gamma_{6M})^2 - 4[\gamma_{6M}^2 - C(\gamma_{6M} - 1)/2]}}{2\gamma_{6M}^2 - C(\gamma_{6M} - 1)} = 0.528$$

Now with the Mach number the static pressure and total pressure can be calculated

$$p_{6M} = p_5 \frac{C_1}{1 + \gamma_{6M} M_{6M}^2} = 108.90 \text{ kPa} \text{ and } P_{t6M} = P_{t5} \cdot \pi_{Mf} = 129.81 \text{ kPa}$$

# Afterburner

$$\eta_{AB} = \frac{(\dot{m}_{air} + \dot{m}_{fuel} + \dot{m}_{AB})h_{t7} - (\dot{m}_{fuel} + \dot{m}_{air})h_{t6M}}{\dot{m}_{AB}(LHV)} \left( \frac{\Delta h_{real}}{\Delta h_{ideal}} \right)$$

$$\Rightarrow f_{AB} = \frac{(1+f)(h_{t7} - h_{t6M})}{(\eta_{ab} \times LHV - h_{t7})} = 0.03850$$

If the loss in the pressure is equivalent to 5%, as in the compressor using the isentropic relations, the expansion can be calculated.

As before, the afterburner is analog to the combustor analysis .

$$P_{t7} = 0.92P_{t6} = 119.426 \text{ kPa}$$

# Nozzle

The first thing to analyze is the nozzle pressure recovery

$$P_{t9} = 0.95 P_{t7} = 0.95 \cdot 119.426 = 113.454 \text{ kPa}$$

Knowing the exit static pressure ratio being  $p_9/p_0 = 3.8$ , gives  $p_9 = 38 \text{ kPa}$ , from there the exit Mach number is calculated

$$M_9 = \sqrt{\frac{2}{\gamma_{AB} - 1} \left[ \left( \frac{P_9}{p_9} \right)^{\frac{\gamma_{AB}-1}{\gamma_{AB}}} - 1 \right]} = 1.3836$$

Static Temperature

$$T_9 = \frac{T_{t9}}{1 + \left( \frac{\gamma_{AB} - 1}{2} M_9^2 \right)} = 1553.84 \text{ K}$$

Speed of Sound

$$a_9 = \sqrt{(\gamma_{AB} - 1) C p_{AB} T_9} = 760.59 \text{ m/s}$$

Jet speed

$$V_9 = M_9 \cdot a_9 = 1052.32 \text{ m/s}$$

Effective velocity

$$V_{9eff} = V_9 + \frac{a_9^2 (1 - p_0/p_9)}{\gamma_{AB} \cdot V_9} = 1363.91 \text{ m/s}$$



# Performance Parameters

Specific Thrust

$$\frac{F_n}{\dot{m}_0(1 + \alpha)a_0} = \left( \frac{1 + \alpha + f + f_{AB}}{1 + \alpha} \right) \frac{V_{9eff}}{a_0} - M_0 = 2.737$$

Thrust Specific Fuel Consumption

$$TSFC = \frac{(f + f_{AB})/(1 + \alpha)a_0}{F_n/\dot{m}_0(1 + \alpha)a_0} = 48.481 \frac{\frac{mg}{s}}{N}$$

Thermal Efficiency

$$\eta_{TH} = \frac{(1 + \alpha + f + f_{AB})V_{9eff}^2 - (1 + \alpha)V_0^2}{2(f \cdot Q_R + f_{AB} \cdot Q_{RAB})} = 47.22 \%$$

Propulsive Efficiency

$$\eta_p = \frac{2F_n V_0}{\dot{m}_0 V_{9eff}^2 - (1 + \alpha)\dot{m}_0 V_0^2} = 62.27 \%$$

Overall Efficiency

$$\eta_{ov} = \eta_{th} \cdot \eta_p = 29.40 \%$$

# Summary

- Various engine configurations and their analysis for steady, one-dimensional flow design- point:
  - Turbojet
  - Turbojet with Afterburner
  - Turbofan with separate exhaust
  - Turbofan with Mixed exhaust
- Included the effects of fluid viscosity and thermal conductivity and other irreversibilities using component efficiencies definitions.